

Gravity as Structural Necessity

An AASC Account of the GR-Type Physical Interior

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Abstract

This paper is explicitly about gravity as seen through the AASC lens. It does not begin by identifying gravity with force, curvature, metric field content, or a quantization target. It defines gravity in the AASC sense as the structural role in a physical theory corresponding to the constraint architecture required for admissible physical construction: standing surface, admissible redescription, observable quotient, constraint closure, and multiplier-type Hamiltonian presentation. The central claim is a typing and forcing theorem, not a numerical derivation of Einstein's equations: once the AASC kernel is specialized to a regular-stratum physical interior with quotient identity, observable completeness, and admissible continuation, the admissible interior must instantiate the role canonically realized by GR-type constrained-Hamiltonian architecture. In the local Hamiltonian regime this means coisotropic standing, characteristic redescription, local first-class constraint generators, and multiplier-type Hamiltonian presentation. These structures are not added as primitive gravitational ontology; they are the constraint form required for an admissible physical interior to exist. A world lacking this role is therefore not an admissible non-gravitational physical interior, but a failure to instantiate the target comparison class. The paper proves the local AER forcing theorem in-manuscript, gives the fixed-domain closure argument excluding same-scope rivals, explains canonical GR as the paradigmatic realization of AASC gravity, and gives a separate fixed-target analysis of projection architecture and primitive GR-to-QM quantization. This result is structural rather than phenomenological: it identifies the conditions under which gravitational theory is possible, not the detailed dynamics of any specific gravitational model. For an independently supplied joint target, the paper constructs canonical role profiles from its standing uses and proves their quotient descent. Structural Unification establishes the paired one-interior result for a determinate joint target. The later Modules A–D capstone proves complete constraint-native gravity–quantum realization of its exact frozen target. This paper records that positive corpus result alongside the explicit native-map and coverage conditions of Theorem 11.4(ii). For a fixed realized pair, it gives the exact condition for a direct projection-to-projection map and excludes an undeclared second foundational authorizer.

Keywords: admissibility; constraint structure; canonical gravity; projection equivalence; quantum gravity foundations; AASC

Roadmap and exact claim

Core thesis. Gravity is not introduced as a primitive force, a primitive metric field, a curvature ontology, or a quantization target. Within AASC, gravity is the admissibility-forced constraint role of the physical interior. More precisely: gravity is fixed by Definition 2.1, and if an admissible physical interior exists in the AER comparison class, then that gravitational role is instantiated as GR-type canonical constraint architecture. A world lacking such structure is not a same-scope non-gravitational admissible world; it fails to instantiate the target. This role includes the zero-constraint endpoint; a particular metric interaction requires further native realization.

The proof order is:

$$\begin{aligned} \text{AASC kernel} &\Rightarrow \text{physical quotient architecture} \\ &\xrightarrow{\text{regular Hamiltonian specialization}} \text{AER interface} \\ &\Rightarrow \text{coisotropic/first-class/multiplier forcing} \\ &\Rightarrow \text{AASC gravity / GR-type interior.} \end{aligned}$$

The local AER forcing theorem is proved here. The wider AASC corpus is used as a dependency source and comparison-class background; Appendix A records exactly which upstream results are invoked and which results are re-proved in this paper.

The specialization arrow fixes the comparison class in Definition 5.1. Kernel governance is necessary for the non-degenerate physical target; the symplectic carrier, regular quotient, and observable compatibility clauses specify the Hamiltonian interface studied here. The proof then derives characteristic equality, constraint closure, and multiplier presentation from that interface. This separates the premises available at each step without making kernel governance optional or importing a metric or quantum realization into the forcing argument. In particular, the quantum target used in Section 11 is not an input to Theorem 8.2.

The paper has four layers. Sections 3–4 fix the target phenomenon and derive the quotient role discipline required for physical standing. Sections 5–7 prove the local Hamiltonian forcing result. Sections 8–10 identify the forced structure as gravity in the AASC sense and close same-scope rivals. Sections 11–12 construct canonical role profiles for a supplied joint target, state the additional native realization obligations, prove the fixed-pair factorization criterion and authority obstruction, and give the constructive certificate program. The joint carrier and native projections are not inferred from the two separately specified targets.

The principal theorem is Theorem 8.2. Its content is structural rather than empirical: it does not fix the numerical gravitational coupling, does not derive the Einstein equations as the only possible dynamical law over all conceivable scopes, and does not prohibit richer-scope theories. It proves that the admissible physical interior cannot lack gravity in the AASC sense while remaining the same target.

Contents

Roadmap and exact claim	2
1 Introduction: gravity is usually mis-typed	3
2 Gravity in the AASC sense	5

3	The target phenomenon and the AASC kernel	6
4	From physical standing to quotient architecture	9
5	AER specialization: regular-stratum observable Hamiltonian evaluation	10
5.1	Regular stratum and intrinsic AER data	11
5.2	AER compatibility conditions	12
6	Coisotropic forcing and characteristic identification	13
6.1	Interface necessity	15
7	Constraint generators, first-class closure, and multiplier form	15
7.1	Local constraint generators	15
7.2	First-class closure	16
7.3	Presentation arbitrariness and multiplier form	17
8	The AASC gravity theorem	19
9	Canonical GR as a downstream realization	22
10	Fixed-domain closure of the GR-type interior	24
11	Projection architecture and primitive quantization	26
12	Positive program: constraint-native gravity	32
13	Non-claims, scope limits, and attack map	34
13.1	Failure modes if GR-type structure is removed	34
13.2	Non-claims	35
13.3	Hostile-referee attack map	35
14	Conclusion	37
A	Dependency ledger and source map	37
B	Minimality witnesses	41
C	Symbol index	42

1 Introduction: gravity is usually mis-typed

The usual starting question for gravity is ontological (for standard constrained-Hamiltonian and quantum-gravity background see [1, 2, 4, 5, 7, 8, 9]): is gravity a force, curvature, a field, an emergent interaction, or an object to be quantized? The AASC framework asks a different question. It asks what structure must exist for a physical domain to support admissibility-bearing, identity-preserving, reusable physical content. On that question gravity is not first typed as a substance or primitive field. It is typed as a constraint role.

The paper does not derive Einstein’s field equations, Newtonian limits, cosmological boundary conditions, or the numerical gravitational coupling. Its claim is upstream: it identifies the gravitational constraint architecture without which an admissible physical interior cannot be well-typed.

The result is therefore not a replacement for GR as empirical physics; it is an account of why the role occupied by GR is structurally necessary inside the AASC target.

The thesis of this paper is:

Gravity (in the role-theoretic sense of Definition 2.1) is the forced constraint role of an admissible physical interior.

The phrase “in the AASC sense” is deliberate and is made explicit in Definition 2.1. It does not mean that this manuscript assumes gravity is identical to curvature or derives the full Einstein field equations from no premises. It means that the constrained-Hamiltonian architecture characteristic of canonical GR—standing surface, quotient physical observables, characteristic/gauge redescription, first-class closure, and multiplier-type presentation dynamics—is the canonical physical realization of the gravity role.

A minimal working understanding of admissibility, standing, and quotient structure is sufficient to follow the argument.

The claim is comparison-class relative. It is not a global classification of every conceivable formal system, every dissipative or non-Hamiltonian model, every Galilean effective system, or every richer-scope theory with additional state variables. The claim concerns the regular-stratum AER physical interior defined below: a Hamiltonian observable-evaluation regime with quotient identity, observable completeness, and admissible continuation. A system outside that target may be coherent, but it is not a same-scope counterexample to the theorem proved here.

The intuitive inversion is simple. Standard physics often says: spacetime exists and physical processes occur in it. The AASC reading says: an admissible physical interior must support determinate same-domain state identity, observables, lawful redescription, and admissible continuation; that demand forces a constraint architecture; the geometric presentation of that architecture is what appears conventionally as gravitational structure. Gravity is therefore not added to an otherwise complete admissible world. It is part of the condition under which such a world can be admissibly presented. Section 2 makes this role-theoretic use of the term precise before the technical argument begins.

Existing approaches attempt to model gravity as a dynamical sector. The present work instead identifies the structural conditions under which gravity must exist at all.

The contribution of this paper is not to describe gravitational structure, but to demonstrate that such structure is necessary for the existence of an admissible physical interior through the forcing chain proved below (Theorems 7.10 and 8.2; see Sections 5–6). This converts gravity from a contingent modeling choice into a structural inevitability.

The results are formulated in a mathematically explicit but conceptually motivated form, intended to bridge foundational physics and formal structure analysis.

This viewpoint explains two otherwise puzzling facts. First, the characteristic structures of canonical gravity—constraints, gauge/redescription, lapse/shift freedom, and Hamiltonian multiplier form—are not defects or merely technical complications. They are exactly what one should expect when physical content is quotient content rather than representation content. Second, a jointly realized physical target fixes the standing content of its gravitational and quantum projections. Section 11 distinguishes constructing that common realization from asking whether one projected output determines the other. A compatible direct map requires a physical-fibre condition; an undeclared second foundational authorizer is excluded by the fixed-target kernel.

Contribution

This paper contributes a gravity-facing bridge theorem inside the AASC corpus. The larger AASC sequence supplies the kernel, fixed-domain, and role-decomposition background [14, 16, 17]. The larger AER and physical-quotient manuscripts supply the constrained-Hamiltonian rigidity lineage and the common anchor/tensor/skin classification. Common role classification and native joint realization carry distinct proof obligations, made explicit here. The present paper isolates the gravitational content of that result and states it in a form suitable for a physics-facing reader: gravity is forced because the admissible interior cannot form without the GR-type constraint role.

The manuscript is audit-grade in the following sense. Every load-bearing step required for the main result is either proved here or listed explicitly as an imported AASC dependency in Appendix A. The local mathematical heart of the argument—coisotropic forcing, characteristic identification, first-class closure, and multiplier form—is proved in Sections 6–7.

2 Gravity in the AASC sense

This paper is explicitly about gravity. The needed clarification is that the term is used role-theoretically rather than by importing one conventional ontology of gravity at the outset. The AASC question is not first “what substance, field, or geometry is gravity?” It is: what structural role must be present for physical content to remain admissible under identity-preserving redescription and continuation?

In this paper, “gravity” is used strictly in the role-theoretic sense defined in Definition 2.1, corresponding to the structural position occupied by canonical gravitational theory in physical models.

Definition 2.1 (AASC gravity). *Gravity in the AASC sense is the structural role in a physical theory corresponding to the constraint architecture required for admissible physical construction: a standing surface, admissible redescription, constraint closure, and multiplier-type Hamiltonian presentation. A theory has gravity in the AASC sense exactly when it instantiates this role in the fixed physical-interior comparison class.*

Remark 2.2 (Role-theoretic use of the word “gravity”). Definition 2.1 does not identify gravity with curvature, force, or a particular field content. It identifies gravity with the admissibility-forced constraint role realized canonically by general relativity. Curvature, metric variables, connection variables, ADM data, and lapse/shift functions are therefore not denied; they are typed as faithful presentations or projections of this role when they preserve the standing-bearing constraint architecture.

In conventional physics, gravity is introduced as a dynamical sector governing spacetime geometry. In the present framework, gravity is not introduced at the foundational level. It is identified as the necessary constraint structure without which physical description cannot be made admissible. What appears as spacetime curvature in GR is, in this sense, the geometric presentation of a deeper admissibility constraint.

Conventional entry point	AASC entry point
Gravity is curvature, force, or metric dynamics.	Gravity is the admissibility-forced constraint role of the physical interior.
Spacetime geometry is treated as primitive or near-primitive.	Quotient identity, standing, and admissible continuation are the load-bearing primitives.
GR describes gravity by giving a geometric field theory.	GR canonically realizes the gravity role through constrained-Hamiltonian structure.
The foundational problem is to quantize gravity.	The foundational problem is to realize the common constraint architecture whose projections are gravitational and quantum.

Proposition 2.3 (Gravity–constraint role identification). *Within the AASC comparison class, the structural role occupied by canonical gravitational theory is exactly the role defined as AASC gravity in Definition 2.1. Any admissible physical theory instantiating this role is therefore gravitational in the AASC sense.*

Proof. Canonical GR instantiates a standing surface, namely its constraint manifold; admissible redescription, namely gauge/diffeomorphism and refoliation orbits; closure, namely the first-class constraint algebra; and continuation, namely Hamiltonian constraint evolution with lapse and shift presentation freedom. By Definition 2.1, these clauses exhaust the AASC gravity role. Any admissible physical theory instantiating these features realizes gravity in the same structural sense. This identifies the shared structural role through its canonical GR realization; claims of GR-specific metric dynamics or gravitational coupling require the corresponding native-realization data. $\square$

Structural necessity and canonical realization. Under the stated AER hypotheses and AER-P, Theorem 8.2 forces coisotropic standing, characteristic redescription, first-class closure, and multiplier presentation. Definitions 2.1 and 8.1 identify this necessary architecture as *AASC gravity*; its necessity is established by the forcing proof. Proposition 2.3 identifies the same structural role in canonical GR. The necessity result includes the zero-constraint endpoint. GR-specific metric dynamics and coupling are established at the level of their native realizations.

3 The target phenomenon and the AASC kernel

This section fixes the target phenomenon. The proof does not begin with a metric, a connection, a Hilbert space, a gauge group, or a Hamiltonian constraint. It begins with the minimal conditions under which a physical construction can be reused as the same determinate content.

Definition 3.1 (Admissible physical interior). *An admissible physical interior is a domain supporting construction or evaluation tokens whose outputs are treated as determinate same-domain physical contents, whose lawful continuation is identity-preserving, whose admissible and inadmissible uses are distinguishable at the point of use, and whose failed uses cannot be retroactively converted into standing without explicit repair or scope change.*

Definition 3.2 (AASC kernel). *The AASC kernel is*

$$K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}),$$

where *Adm* distinguishes licensed from unlicensed continuation, *Stand* is reusable act-status, *Ref* fixes the same-domain target, and *Irr* blocks same-act post-hoc repair of a failed act.

Lemma 3.3 (Admissibility is necessary). *If a physical domain lacks a boundary distinguishing licensed from unlicensed continuation at the point of use, it does not instantiate an admissible physical interior.*

Proof. An output is evaluable only if some continuations count as licensed for that output and others do not. If every continuation is equally licensed, the output carries no governance content. If no continuation is assessable, the output has no admissibility status. In either case the domain may contain transitions or traces, but not admissibility-bearing physical construction. $\square$

Lemma 3.4 (Standing is necessary). *If outputs of physical construction cannot acquire reusable act-status as licit same-domain content, the domain does not instantiate an admissible physical interior.*

Proof. The target requires later use of the output as the same content. Without standing, the output is merely an event, inscription, or transient state; it supplies no licit downstream premise, state, observable, or record. Reuse as physical content is therefore impossible. $\square$

Lemma 3.5 (Reference is necessary). *If physical construction outputs do not fix a determinate object of same-domain continuation, the domain does not instantiate an admissible physical interior.*

Proof. Identity-preserving continuation requires a fact of the matter about which object is being continued. Without reference, a later operation cannot be distinguished from replacement, relabelling, or change of target. Thus same-domain physical identity is not defined. $\square$

Lemma 3.6 (Irreversibility is necessary). *If an inadmissible act can be retroactively converted into admissible standing on the same fixed domain without repair or scope change, the domain does not instantiate an admissible physical interior.*

Proof. The target requires that admissibility status be fixed at the relevant use boundary. If a failed act can later be reauthored as successful while preserving the same act and same domain, then failure and success were not determinate statuses of the original construction. The object has been reset rather than continued. $\square$

Theorem 3.7 (Kernel necessity). *Every nondegenerate admissible physical interior instantiates the AASC kernel $K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr})$, up to same-domain governance equivalence. Each component is minimal: removing any one component destroys the target rather than producing a weaker faithful realization of it.*

Proof. Let a domain instantiate Definition 3.1. It must distinguish licensed from unlicensed continuation, or Lemma 3.3 applies. Its outputs must be reusable as same-domain content, or Lemma 3.4 applies. It must fix determinate objects of continuation, or Lemma 3.5 applies. It must block same-act post-hoc repair, or Lemma 3.6 applies. Hence every nondegenerate realization instantiates all four components.

Minimality follows by the same four lemmas. Removing admissibility yields unassessability; removing standing yields non-reuse; removing reference yields indeterminate objecthood; removing irreversibility yields retrospective reset. These are distinct collapse modes, and each destroys the target phenomenon. $\square$

Corollary 3.8 (Typed denial burden). *A denial of the kernel has one of three forms: derivation from a governance-free lower layer, faithful same-domain counterexample, or rejection/alteration of the target phenomenon. The first two are excluded for the target of Definition 3.1; the third is a scope change rather than a same-scope denial.*

Proof. A same-domain denial preserving the target must either show that the missing governance work is supplied from below or show that the target occurs without the missing component. The latter contradicts Theorem 3.7; the former presupposes admissible derivation, standing-bearing outputs, reference-fixed target, and irreversibility of failure, and hence already uses governance-equivalent structure. If the denial instead rejects reusable same-domain physical content, it has changed the target. $\square$

Definition 3.9 (Regime qualification). *A regime qualification for a proposed physical theorem object is the list of structural conditions under which the object is claimed to be evaluable as the same object across redescription, continuation, and reuse. In this paper the qualifications are introduced in layers: first the kernel of Definition 3.2, then quotient architecture, then the regular-stratum AER Hamiltonian interface.*

Theorem 3.10 (Minimal regime qualification; anti-circularity). *The conditions used to define the admissible physical interior are not premises encoding GR-type structure. They are minimal target-qualification conditions for identity-bearing, reusable, proposition-bearing physical content. Removing any kernel condition yields loss of evaluability, reuse, determinate target identity, or irreversible commitment. Adding GR-specific features such as a metric, ADM variables, Einstein equations, constraint functions, lapse, shift, or first-class closure is not part of Definition 3.1 and enters only after the AER specialization and local forcing proof.*

Proof. The kernel clauses contain no metric, connection, curvature tensor, ADM phase space, constraint algebra, lapse, shift, or multiplier Hamiltonian. By Theorem 3.7, they are target-enabling conditions for reusable same-domain physical content. The collapse of each removed clause is established by Lemmas 3.3–3.6. The GR-type clauses are conclusions of later results: quotient necessity is proved in Theorem 4.4; coisotropy is proved in Theorem 6.3; first-class closure is proved in Theorem 7.5; multiplier form is proved in Theorem 7.8. Since the alleged GR-specific content appears as a conclusion rather than as a definitional premise, the argument is not circular in the relevant sense. $\square$

Corollary 3.11 (Faithful counterexample burden). *A faithful counterexample to the main theorem must preserve the admissible physical interior, preserve the regular-stratum AER comparison class, and then exhibit a non-GR-type same-scope realization. A construction that rejects reusable same-domain physical content, leaves Hamiltonian observable evaluation, weakens observable completeness, or adds new state-level anchors is a scope change rather than a same-scope counterexample.*

Proof. The target is fixed by Definitions 3.1 and 3.9. If a proposed counterexample does not instantiate that target, it does not contradict a theorem about that target. If it instantiates the target but alters the later AER comparison class, it is not a same-scope rival to the derived AER object. If it preserves both, then it must survive the local forcing theorems and the fixed-domain closure theorem; Sections 6–10 show that it cannot. $\square$

Remark 3.12 (Why this is not a philosophy premise). The kernel theorem is not a definition of AASC vocabulary into truth. It is a necessity statement for a typed use: determinate physical content reused under identity-preserving continuation. Systems that decline this use are coherent as different systems, but they are not counterexamples to the admissible physical interior.

4 From physical standing to quotient architecture

The AASC kernel by itself is not yet gravity. Gravity appears only after the kernel is specialized to physical content: states, observables, redescription, and continuation. The first consequence is quotient architecture. Physical content cannot be identified with a single description, because lawful changes of coordinates, gauge, basis, representative, or presentation may preserve the same physical object.

Definition 4.1 (Physical presentation). *A physical presentation X consists of a state carrier P_X , a class $\mathcal{O}_X$ of outcome-relevant observables, a lawful redescription relation $\sim_X$, and an admissible continuation class Cont_X . Its physical quotient is $P_X / \sim_X$, when this quotient exists in the required regularity class.*

Definition 4.2 (Admissibility type). *The admissibility type of a presentation X is*

$$\text{Typ}(X) = (\text{Anc}_X, \text{Ten}_X, \sim_X, \text{Cont}_X),$$

where anchors fix domain, boundary, regime, and admissibility envelope; tensors are standing-bearing witnesses; $\sim_X$ is lawful identity-preserving equivalence; and Cont_X is admissible continuation/composition.

Definition 4.3 (Same-domain well-typed map). *A map $F : X \rightarrow Y$ is same-domain well typed when it preserves admissibility type up to lawful role-preserving equivalence:*

$$F_*(\text{Anc}_X) = \text{Anc}_Y, \quad F_*(\text{Ten}_X) = \text{Ten}_Y, \quad x \sim_X x' \Rightarrow F(x) \sim_Y F(x'), \quad F_*(\text{Cont}_X) = \text{Cont}_Y.$$

Failure of one of these clauses is anchor motion, tensor-witness change, quotient motion, or continuation motion.

Theorem 4.4 (Quotient necessity for physical standing). *If a physical domain supports reusable same-domain physical content under lawful redescription, then its standing-bearing content factors through a quotient by admissible redescription.*

Proof. Lawful redescription preserves the physical content being reused. A standing-bearing evaluation that assigned different contents solely to two lawfully equivalent representatives would therefore privilege a representative or introduce an additional standing-bearing discriminator. The former violates reference stability; the latter must be declared as a change of physical content or admissibility type. Thus every admitted standing-bearing evaluation is invariant under the lawful identity-preserving relation.

Let $q_X : P_X \rightarrow P_X / \sim_X$ be the set quotient and let $u : P_X \rightarrow Z$ be such an evaluation. Define

$$\bar{u}([p]) := u(p).$$

Invariance makes this definition independent of the representative. Hence $u = \bar{u} \circ q_X$, and surjectivity of q_X makes $\bar{u}$ unique. The same argument applies to a joint signature of admitted standing-bearing uses. This proves the necessary factorization.

Factorization alone does not assert that the admitted witnesses separate distinct quotient classes, or that the set quotient has a smooth manifold structure. The smooth quotient submersion and observable separation used by the local Hamiltonian proof are supplied by the AER specialization in Section 5. $\square$

Definition 4.5 (Admissibility-bearing quotient architecture). *An admissibility-bearing quotient architecture is a role-labelled object*

$$\mathcal{A} = (\text{Anc}_{\mathcal{A}}, \text{Ten}_{\mathcal{A}}, \text{Skin}_{\mathcal{A}}, \sim_{\mathcal{A}}, \text{Cont}_{\mathcal{A}}),$$

where anchors fix the domain and admissibility envelope, tensors carry invariant standing-bearing content, skin is representational surplus, $\sim_{\mathcal{A}}$ is lawful identity-preserving equivalence, and $\text{Cont}_{\mathcal{A}}$ is admissible continuation.

Definition 4.6 (Projection). Write $\mathcal{A}$ also for the presentation carrier of an admissibility-bearing architecture. A role projection to a declared target X is a map $\Pi_X : \mathcal{A} \rightarrow X$ preserving lawful equivalence, hence inducing

$$\bar{\Pi}_X : \mathcal{A}_{\text{phys}} \rightarrow X_{\text{phys}}, \quad \bar{\Pi}_X([a]) = [\Pi_X(a)],$$

where the subscripts denote physical quotients. Its claimed physical output is its realized image unless coverage of the full declared target is proved. It must preserve the standing required by its declared role; Definition 11.1 specifies the complete architectural conditions. A physical fibre of one role projection may contain distinctions that remain standing-bearing for another role. Such a fibre is not thereby global skin.

Theorem 4.7 (No fourth same-domain role). *Within a fixed admissibility-bearing physical presentation, every descriptive element is anchor, tensor, or skin. No fourth same-domain standing-bearing role survives.*

Proof. Take a descriptive element e . If varying e changes the domain, boundary, admissibility envelope, regime, or well-typedness conditions, then e is anchor. Hold anchors fixed. If varying e changes some invariant physical observable, record, constraint witness, compatibility relation, or admissible continuation output, then e is tensor. If varying e changes neither the anchors nor standing-bearing witness/continuation content, then no admissible downstream use can distinguish it; it is skin.

A purported fourth role would have to carry standing-bearing force while changing neither domain nor invariant content. That is precisely hidden unwitnessed structure or a second same-scope classifier, both excluded by Theorem 3.7 and Theorem 4.4. Hence the trichotomy is exhaustive. $\square$

Corollary 4.8 (No second same-scope classifier). *A selector, canonical representative, preferred foliation, basis choice, gauge choice, measure, or branch rule cannot act as an additional same-scope authorizer. If it changes anchors it is scope motion; if it changes tensors it is declared new load; if it changes neither it is skin.*

Proof. Apply Theorem 4.7 to the proposed selector. Same-scope authority without anchor or tensor effect is impossible. With anchor or tensor effect it is no longer an undeclared same-scope selector. $\square$

5 AER specialization: regular-stratum observable Hamiltonian evaluation

We now specialize the quotient architecture to the local Hamiltonian setting. The aim is not to assume GR. The aim is to show that once admissible physical standing is evaluated through Hamiltonian observables on a regular stratum, the GR-type constrained architecture is forced.

Kernel roles and their AER representation. The kernel governs determinate, reusable physical content. The following table identifies how the stated Hamiltonian specialization represents its roles. The smooth carrier, symplectic form and observable bracket are declared specialization data; the table does not derive them from the kernel alone.

Kernel role	Representation in the declared AER interface
Adm	The fixed admissibility envelope and the allowed standing-preserving continuations.
Stand	State standing represented by membership in the smooth surface $\mathcal{C}$; observable uses must satisfy the declared compatibility clauses.
Ref	Identity-preserving redescription represented by π , with physical observables constant on its classes.
Irr	A failed use cannot acquire standing by same-act post-hoc reclassification; repair or scope change must be explicit.

The last row concerns the status of a use. It permits reversible Hamiltonian evolution and lawful returns to an earlier state or quotient class; it supplies no thermodynamic arrow of time. Coisotropy and characteristic equality will follow from the stated AER data and observable compatibility conditions; multiplier form also uses the separately stated AER-P.

Definition 5.1 (AER comparison class). *The AER comparison class in this manuscript consists of regular-stratum Hamiltonian observable-evaluation presentations satisfying the following target restrictions: a symplectic state carrier is fixed locally; admissible states form a smooth standing surface; lawful redescription is represented by a smooth quotient submersion; physical observables are quotient functions; observable Poisson brackets descend extension-independently; observables separate quotient-transverse directions; and no nonzero presymplectic class remains invisible to all physical observables. Presentation-parametrized evolution is added only when multiplier-type gravitational dynamics is being classified.*

Remark 5.2 (Comparison-class boundary). Definition 5.1 is a scope guard. Dissipative systems, non-Hamiltonian models, nonregular singular quotients, Galilean effective regimes, and richer theories with additional state variables are not ruled out by this paper. They are outside the same theorem object unless they supply a transport-complete certificate preserving the AER clauses. The theorem proved below is therefore structural necessity inside this comparison class, not a universal taxonomy of all possible physical formalisms.

5.1 Regular stratum and intrinsic AER data

Let (S, ω) be a finite-dimensional symplectic manifold. All results are local on a regular stratum: submanifolds have constant codimension, quotient maps are smooth submersions, and local pseudogroups act smoothly.

Definition 5.3 (Standing surface). *A standing surface is a smooth embedded submanifold $\mathcal{C} \subset S$. Binary standing is membership in $\mathcal{C}$:*

$$\text{Stand}(x) = 1 \iff x \in \mathcal{C}.$$

Definition 5.4 (Admissible redescription pseudogroup). *A local pseudogroup $\mathcal{G}$ on $\mathcal{C}$ is a collection of local diffeomorphisms closed under composition and inversion whenever defined. It generates the equivalence relation*

$$x \sim y \iff \exists g \in \mathcal{G} \ y = g(x).$$

Assume the local quotient $S_{\text{phys}} = \mathcal{C} / \sim$ exists and that $\pi : \mathcal{C} \rightarrow S_{\text{phys}}$ is a smooth submersion.

Definition 5.5 (Physical observable). *A physical observable is a smooth function $O : \mathcal{C} \rightarrow \mathbb{R}$ constant on $\sim$ -classes. Equivalently, $O = \tilde{O} \circ \pi$ for some smooth $\tilde{O} : S_{\text{phys}} \rightarrow \mathbb{R}$.*

Definition 5.6 (Redescription distribution). *The redescription distribution is*

$$\mathcal{D}_x := \ker(d\pi_x) \subset T_x\mathcal{C}.$$

Lemma 5.7 (Vertical involutivity). *The distribution $\mathcal{D} = \ker(d\pi)$ is involutive. Its local leaves are connected components of the fibres of π , hence local admissible-redescription classes.*

Proof. Because π is a smooth submersion, $\mathcal{D}$ is the vertical subbundle of π . If X, Y are vertical vector fields, then they are π -related to the zero vector field on S_{phys} . Naturality of Lie brackets for related vector fields gives $d\pi([X, Y]) = [0, 0] = 0$. Hence $[X, Y]$ is vertical. The fibre statement is the standard submersion theorem. $\square$

The observable class used here. Definition 5.5 includes all smooth quotient functions, pulled back by π . On an ordinary smooth quotient manifold these functions separate points, and their differentials span the quotient cotangent space. In a quotient chart, coordinate functions with a smooth cutoff realize any prescribed cotangent vector at its centre; pullback by the submersion identifies that space with $\text{Ann}(\ker d\pi_x)$. Thus the named AER-Sep clause below is explicit in the proof ledger even where the full smooth observable class already supplies it. No restricted observable algebra is substituted in this manuscript.

5.2 AER compatibility conditions

Let $\iota : \mathcal{C} \hookrightarrow S$ be inclusion and $\omega_{\mathcal{C}} := \iota^*\omega$. For $x \in \mathcal{C}$, write

$$(T_x\mathcal{C})^{\perp_{\omega}} := \{\xi \in T_xS : \omega_x(\xi, w) = 0 \ \forall w \in T_x\mathcal{C}\}.$$

Axiom 5.8 (AER-Obs_{min}: extension-independent Poisson descent). *For any two physical observables $O = \tilde{O} \circ \pi$ and $P = \tilde{P} \circ \pi$, choose arbitrary smooth extensions $\overline{O}, \overline{P}$ to a neighbourhood of $\mathcal{C}$ in S . The restriction $\{\overline{O}, \overline{P}\}|_{\mathcal{C}}$ is independent of the chosen extensions and depends only on the quotient class: there exists a bracket $\{\cdot, \cdot\}_{\text{phys}}$ on S_{phys} such that*

$$\{\overline{O}, \overline{P}\}|_{\mathcal{C}} = \{\tilde{O}, \tilde{P}\}_{\text{phys}} \circ \pi.$$

In particular, if one restricted observable is the zero observable on $\mathcal{C}$, its bracket with any physical observable restricts to zero on $\mathcal{C}$.

Axiom 5.9 (AER-Sep: quotient separation and cotangent spanning). *On the regular stratum: (i) if $\pi(x) \neq \pi(y)$, some physical observable separates x and y ; and (ii) for each $x \in \mathcal{C}$, the differentials of physical observables span $\text{Ann}(\mathcal{D}_x) \subset T_x^*\mathcal{C}$:*

$$\text{Ann}(\mathcal{D}_x) = \text{span}\{dO_x : O \text{ physical}\}.$$

Axiom 5.10 (AER-Max: maximal same-scope observability). *For each $x \in \mathcal{C}$ and each $v \in T_x\mathcal{C}$, if $v \notin \ker(\omega_{\mathcal{C},x})$, then there exists a physical observable O with $dO_x(v) \neq 0$. Equivalently, physical observables separate nonzero classes of the presymplectic quotient $T_x\mathcal{C}/\ker(\omega_{\mathcal{C},x})$.*

Remark 5.11 (Status of AER-Max). AER-Max is the observable-completeness clause. It does not assume reduced symplectic geometry or $\mathcal{D} = \ker(\omega_{\mathcal{C}})$. It excludes only hidden same-scope Hamiltonian directions: nonzero presymplectic classes invisible to every physical observable.

The content of observable maximality. Under AER-Sep, AER-Max is equivalent to the inclusion $\mathcal{D}_x \subseteq \ker(\omega_{\mathcal{C},x})$. The forward implication is proved in Lemma 6.2. Conversely, if this inclusion holds and $v \notin \ker(\omega_{\mathcal{C},x})$, then $v \notin \mathcal{D}_x$. Some covector in $\text{Ann}(\mathcal{D}_x)$ detects v ; since observable differentials span that annihilator, at least one physical observable detects v . This is AER-Max. It supplies one inclusion, while characteristic capture supplies the other. Proposition B.1 shows that the first inclusion cannot be obtained from descent and separation alone.

The quotient-class wording in AER-Max is read within this combined interface. Once Lemma 6.1 has given $\ker(\omega_{\mathcal{C}}) \subseteq \mathcal{D}$, observable differentials annihilate $\ker(\omega_{\mathcal{C}})$, so they define covectors on the presymplectic quotient. AER-Max then says precisely that they detect every nonzero quotient tangent class.

Definition 5.12 (AER physical interior). *An AER physical interior is the regular-stratum data $(S, \omega, \mathcal{C}, \pi, \mathcal{D}, \mathcal{O}_{\text{phys}})$ satisfying Definitions 5.3–5.6 and Axioms 5.8–5.10.*

6 Coisotropic forcing and characteristic identification

This section proves the local theorem that makes the gravitational reading possible. Constraint structure is not assumed. It is forced from extension-independent observable Poisson descent, quotient separation, and observable completeness.

Lemma 6.1 (Characteristic directions are captured by admissible redescription). *Assume AER-Obs_{min} and AER-Sep. Then for every $x \in \mathcal{C}$,*

$$(T_x\mathcal{C})^{\perp\omega} \subseteq \mathcal{D}_x.$$

Consequently $\mathcal{C}$ is coisotropic.

Proof. Fix $x \in \mathcal{C}$ and $\xi \in (T_x\mathcal{C})^{\perp\omega}$. The covector $\alpha := \omega_x(\xi, \cdot)$ vanishes on $T_x\mathcal{C}$. Choose a submanifold chart (y, z) centred at x with $\mathcal{C} = \{z = 0\}$. Since α is conormal, write $\alpha = \sum_a a_a dz_x^a$, and set $H = \sum_a a_a z^a$ locally. Then $H|_{\mathcal{C}} = 0$ and $dH_x = \alpha$. With the convention $\omega(X_H, \cdot) = dH$, nondegeneracy gives $X_H(x) = \xi$.

Let O be any physical observable and $\bar{O}$ any extension. Since $H|_{\mathcal{C}} = 0$ is the zero physical observable, AER-Obs_{min} gives

$$\{H, \bar{O}\}|_{\mathcal{C}} = 0.$$

At x , this says $d\bar{O}_x(X_H(x)) = d\bar{O}_x(\xi) = 0$.

We must first show that $\xi \in T_x\mathcal{C}$. Let K be any smooth function with $K|_{\mathcal{C}} = 0$. Applying the same zero-observable clause to (H, K) gives $dK_x(\xi) = 0$. Differentials of functions vanishing on $\mathcal{C}$ span $\text{Ann}(T_x\mathcal{C})$; hence ξ annihilates $\text{Ann}(T_x\mathcal{C})$, so $\xi \in T_x\mathcal{C}$.

Now ξ is tangent to $\mathcal{C}$, so $dO_x(\xi) = d\bar{O}_x(\xi)$, independently of the extension. Hence $dO_x(\xi) = 0$ for every physical observable O . By AER-Sep, differentials of physical observables span $\text{Ann}(\mathcal{D}_x)$. Hence ξ annihilates $\text{Ann}(\mathcal{D}_x)$. Since $\mathcal{D}_x \subset T_x\mathcal{C}$ is finite-dimensional, $\text{Ann}(\text{Ann}(\mathcal{D}_x)) = \mathcal{D}_x$, and therefore $\xi \in \mathcal{D}_x$.

Because $\mathcal{D}_x \subset T_x\mathcal{C}$, the inclusion $(T_x\mathcal{C})^{\perp\omega} \subseteq T_x\mathcal{C}$ holds. This is precisely coisotropy of $\mathcal{C}$. $\square$

Lemma 6.2 (No hidden presymplectic remainder). *Assume AER-Max. Then*

$$\mathcal{D}_x \subseteq \ker(\omega_{\mathcal{C},x})$$

for every $x \in \mathcal{C}$.

Proof. Let $v \in \mathcal{D}_x$. If $v \notin \ker(\omega_{\mathcal{C},x})$, AER-Max supplies a physical observable $O = \tilde{O} \circ \pi$ with $dO_x(v) \neq 0$. But $v \in \ker(d\pi_x)$, so

$$dO_x(v) = d\tilde{O}_{\pi(x)}(d\pi_x(v)) = 0,$$

a contradiction. Therefore $v \in \ker(\omega_{\mathcal{C},x})$. $\square$

Theorem 6.3 (Coisotropic forcing and characteristic identification). *Assume AER-Obs_{min}, AER-Sep, and AER-Max. Then*

$$\mathcal{D} = \ker(d\pi) = \ker(\omega_{\mathcal{C}}).$$

In particular, $\mathcal{C}$ is coisotropic and admissible redescription coincides with the characteristic foliation of $\omega_{\mathcal{C}}$.

Proof. By Lemma 6.1, $\mathcal{C}$ is coisotropic and $(TC)^{\perp\omega} \subseteq \mathcal{D}$. For a coisotropic submanifold, $\ker(\omega_{\mathcal{C}}) = (TC)^{\perp\omega}$. Hence $\ker(\omega_{\mathcal{C}}) \subseteq \mathcal{D}$. Lemma 6.2 gives $\mathcal{D} \subseteq \ker(\omega_{\mathcal{C}})$. Therefore equality holds. The characteristic foliation is by definition the foliation tangent to $\ker(\omega_{\mathcal{C}})$, which now equals the admissible redescription distribution. $\square$

Corollary 6.4 (Reduced symplectic form). *Under Theorem 6.3, there is a unique local symplectic form ω_{phys} on S_{phys} such that*

$$\pi^*\omega_{\text{phys}} = \omega_{\mathcal{C}}.$$

The descended bracket of AER-Obs_{min} is the Poisson bracket induced by ω_{phys} .

Proof. The form $\omega_{\mathcal{C}}$ is closed because ω is closed. Work first in a submersion chart $\pi^{-1}(V) \supset W \simeq V \times F$, with connected local fibre slices. For every vertical vector field Z , Theorem 6.3 gives $\iota_Z\omega_{\mathcal{C}} = 0$. Cartan's formula then gives

$$\mathcal{L}_Z\omega_{\mathcal{C}} = d(\iota_Z\omega_{\mathcal{C}}) + \iota_Zd\omega_{\mathcal{C}} = 0.$$

Thus the form is horizontal and constant along these slices, and descends to a closed form ω_V on V . A quotient tangent vector pairing trivially with all others has a lift in $\ker\omega_{\mathcal{C}} = \ker d\pi$, so it is zero. Hence ω_V is nondegenerate.

To identify its bracket, let $O = \pi^*f$ and $P = \pi^*g$, and choose ambient extensions. If K vanishes on $\mathcal{C}$, the zero-observable clause of AER-Obs_{min} implies $dK(X_{\overline{O}}) = 0$ on $\mathcal{C}$. Thus $X_{\overline{O}}$ is tangent to $\mathcal{C}$. For $w \in TC$,

$$\omega_{\mathcal{C}}(X_{\overline{O}}, w) = dO(w) = df(d\pi(w)).$$

Nondegeneracy of ω_V gives $d\pi(X_{\overline{O}}) = X_f$, and therefore

$$\{\overline{O}, \overline{P}\}|_{\mathcal{C}} = dP(X_{\overline{O}}) = \{f, g\}_{\omega_V} \circ \pi.$$

This is the single quotient bracket specified by AER-Obs_{min}.

If two local lifts of a quotient neighbourhood lie in different fibre components, their induced brackets still agree with that same bracket. AER-Sep supplies cotangent spanning, so equality on physical observables identifies their Poisson bivectors and hence their inverse symplectic forms. The local forms consequently agree on overlaps and define the asserted local ω_{phys} . Uniqueness follows from injectivity of pullback by a submersion. $\square$

Local leaves and quotient identity. The equality of distributions identifies connected local redescription leaves with characteristic leaves. It does not reconstruct every discrete identification of a global quotient from its tangent distribution. The map π , including any such identifications, remains part of the fixed target; the preceding bracket argument checks consistency wherever its quotient charts are compared.

6.1 Interface necessity

Proposition 6.5 (Necessity profile of the Stage II interface). *The three AER compatibility clauses have distinct necessity roles. Failure of AER-Obs_{min} destroys extension-independent quotient-level Poisson closure. Failure of AER-Sep destroys quotient-transverse detectability. Failure of AER-Max leaves a nonzero presymplectic same-scope remainder invisible to all physical observables.*

Proof. If AER-Obs_{min} fails, the bracket of physical observables either depends on ambient extensions or does not descend to S_{phys} ; then the observable algebra is not defined on quotient data. If AER-Sep fails, either two quotient classes are not separated by observables or some covector transverse to $\mathcal{D}$ is invisible to every observable differential; then the quotient is not complete as a physical classifier. If AER-Max fails, there is $v \in T_x\mathcal{C} \setminus \ker(\omega_{\mathcal{C},x})$ with $dO_x(v) = 0$ for all physical O . This is a nonzero presymplectic class carrying Hamiltonian tangent content while remaining invisible to the physical observable algebra. These are respectively closure, separation, and maximality failures. $\square$

7 Constraint generators, first-class closure, and multiplier form

The previous section derived the characteristic structure. We now show that the usual constrained Hamiltonian machinery follows locally.

Hamiltonian sign convention. Throughout the local proof, $\omega(X_f, \cdot) = df$ and $\{f, g\} := dg(X_f) = X_f(g)$. With this paired convention the commutator identity below has the displayed positive sign.

7.1 Local constraint generators

Theorem 7.1 (Coisotropic local normal form). *Let (S, ω) be symplectic and let $\mathcal{C} \subset S$ be a regular coisotropic submanifold of codimension k . For each $x \in \mathcal{C}$, there are local Darboux coordinates $(q^1, \dots, q^n, p_1, \dots, p_n)$ on a neighbourhood U of x such that*

$$\mathcal{C} \cap U = \{p_1 = \dots = p_k = 0\}.$$

Moreover the characteristic distribution $\ker(\omega_{\mathcal{C}})$ is locally spanned on $\mathcal{C} \cap U$ by $X_{p_1}, \dots, X_{p_k}$.

Proof. Cattaneo and Zambon [20, Proposition 9.6] give adapted split Poisson coordinates in which the coisotropic submanifold is defined by $p_1 = \dots = p_k = 0$. Their conormal-rank condition is automatic here because the ambient symplectic form is nondegenerate. The remaining Poisson block is then nondegenerate, and Darboux coordinates on that block complete the claimed symplectic chart. The symplectic dual of the conormal span $\text{span}\{dp_1, \dots, dp_k\}$ is $\text{span}\{X_{p_1}, \dots, X_{p_k}\}$; for a coisotropic submanifold this is exactly the characteristic distribution. This external local normal form is used only after Theorem 6.3 has established coisotropy. $\square$

Definition 7.2 (Local constraint generators). *On a neighbourhood U as in Theorem 7.1, define*

$$\phi_a := p_a, \quad a = 1, \dots, k.$$

Then $\mathcal{C} \cap U = \{\phi_1 = \dots = \phi_k = 0\}$.

Proposition 7.3 (Constraint generators span admissible redescription). *On $\mathcal{C} \cap U$,*

$$\mathcal{D} = \text{span}\{X_{\phi_1}, \dots, X_{\phi_k}\}.$$

Proof. By Theorem 6.3, $\mathcal{D} = \ker(\omega_{\mathcal{C}})$. By Theorem 7.1, $\ker(\omega_{\mathcal{C}})$ is locally spanned by $X_{p_a} = X_{\phi_a}$. $\square$

7.2 First-class closure

Lemma 7.4 (Hamiltonian commutator identity). *For smooth functions f, g on a symplectic manifold,*

$$[X_f, X_g] = X_{\{f, g\}}.$$

Proof. This is the standard identity expressing that Hamiltonian vector fields represent the Poisson algebra as derivations. It follows by applying Cartan's formula to $\iota_{[X_f, X_g]}\omega = \mathcal{L}_{X_f}\iota_{X_g}\omega - \iota_{X_g}\mathcal{L}_{X_f}\omega$, using $d\omega = 0$ and $\mathcal{L}_{X_f}\omega = 0$. $\square$

Theorem 7.5 (First-class closure). *Let ϕ_a be local constraint generators as in Definition 7.2. Then locally*

$$\{\phi_a, \phi_b\} = \sum_{c=1}^k C_{ab}^c \phi_c$$

for smooth coefficient functions C_{ab}^c . Equivalently, $\{\phi_a, \phi_b\} \approx 0$ on $\mathcal{C}$.

Proof. By Proposition 7.3, X_{ϕ_a} is tangent to $\mathcal{C}$ along $\mathcal{C}$. Since ϕ_b vanishes on $\mathcal{C}$, its differential annihilates every vector tangent to $\mathcal{C}$. Therefore, on $\mathcal{C}$,

$$\{\phi_a, \phi_b\} = d\phi_b(X_{\phi_a}) = 0.$$

Because $\phi_1, \dots, \phi_k$ are regular defining functions, they may be used as normal coordinates z in a product chart (y, z) with $\mathcal{C} = \{z = 0\}$. After shrinking the chart, every smooth F vanishing on $\mathcal{C}$ satisfies

$$F(y, z) = \sum_c z^c \int_0^1 \frac{\partial F}{\partial z^c}(y, tz) dt.$$

Applying this identity to $F = \{\phi_a, \phi_b\}$ gives the smooth coefficients C_{ab}^c . For the adapted generators $\phi_a = p_a$, these brackets vanish identically, so the coefficients can be zero. The ideal-membership argument also explains why a different regular constraint presentation remains first class. $\square$

Local invariance under a regular constraint presentation. Fix the ambient symplectic structure, standing surface $\mathcal{C}$, quotient π , physical observables, and continuation target. Suppose two length- k tuples $\phi = (\phi_a)$ and $\psi = (\psi_a)$ regularly define the same codimension- k surface near a point of $\mathcal{C}$. In local coordinates transverse to $\mathcal{C}$, Hadamard factorization gives a smooth matrix M with $\psi = M\phi$. Along $\mathcal{C}$, $d\psi = M d\phi$; since both differential tuples are conormal frames, $M|_{\mathcal{C}}$ is

invertible. Shrinking the neighbourhood makes M invertible throughout it. The product rule then yields

$$X_{\psi_a}|_{\mathcal{C}} = \sum_b M_a^b X_{\phi_b}|_{\mathcal{C}}, \quad \text{span}\{X_{\psi_a}|_{\mathcal{C}}\} = \text{span}\{X_{\phi_a}|_{\mathcal{C}}\} = \mathcal{D}.$$

For the same extension $\widehat{h}$, the very same Hamiltonian has

$$H = \widehat{h} + u^\top \phi = \widehat{h} + v^\top \psi, \quad v = M^{-\top} u.$$

This is local presentation invariance of the characteristic generators and multiplier coefficients for one fixed target. It does not compare different standing surfaces, quotient identifications, observable brackets, or physical continuation laws.

Both tuples here contain exactly $k = \text{codim } \mathcal{C}$ independent generators. Redundant lists need not be related by an invertible matrix, and this local argument supplies no global trivialization of the conormal bundle. First-class closure permits smooth structure functions; constancy of the coefficients is not required.

Corollary 7.6 (Nontrivial redescription excludes second-class standing). *If $\omega_{\mathcal{C}}$ is nondegenerate, then $\mathcal{D} = 0$ and admissible redescription is trivial. Conversely, nontrivial admissible redescription forces degeneracy of $\omega_{\mathcal{C}}$.*

Proof. Theorem 6.3 gives $\mathcal{D} = \ker(\omega_{\mathcal{C}})$. If $\omega_{\mathcal{C}}$ is nondegenerate then $\ker(\omega_{\mathcal{C}}) = 0$, hence $\mathcal{D} = 0$. Conversely $\mathcal{D} \neq 0$ implies $\ker(\omega_{\mathcal{C}}) \neq 0$. $\square$

7.3 Presentation arbitrariness and multiplier form

Axiom 7.7 (AER-P: presentation-parametrized evolution). *There is a family of Hamiltonians $\{H_\alpha\}_{\alpha \in A}$ such that: (i) each flow preserves $\mathcal{C}$ and descends to S_{phys} ; (ii) all presentations induce the same reduced Hamiltonian vector field on S_{phys} , up to dynamically inert reduced Casimirs or constants; and (iii) for all $x \in \mathcal{C}$,*

$$(X_{H_\alpha} - X_{H_\beta})(x) \in \mathcal{D}_x.$$

Representative convention. In the trivial-reduced-evolution clause below, choosing the reduced Hamiltonian to be represented by zero also chooses the presentation Hamiltonian modulo the corresponding dynamically inert constant, as allowed by AER-P(ii). The final pure-multiplier display uses that normalized representative. For an unshifted Hamiltonian the constant term must be retained; weak equality on $\mathcal{C}$ alone cannot remove it. The proof writes the two representatives separately.

Theorem 7.8 (Multiplier-type Hamiltonian rigidity). *Assume AER-P. For each presentation α , locally near $\mathcal{C} \cap U$ there are a reduced Hamiltonian $h_\alpha \in C^\infty(S_{\text{phys}})$, a smooth extension $\widehat{h}_\alpha$ of $\pi^* h_\alpha$, and coefficient functions u_α^a such that*

$$H_\alpha = \widehat{h}_\alpha + \sum_a u_\alpha^a \phi_a.$$

If the presentation family has trivial reduced evolution, then h_α may be represented by zero and

$$H_\alpha = \sum_a u_\alpha^a \phi_a$$

locally, modulo weak equality on $\mathcal{C}$.

Proof. By AER-P(i), X_{H_α} is tangent to $\mathcal{C}$. For any $v \in \mathcal{D} = \ker(\omega_{\mathcal{C}})$,

$$d(H_\alpha|_{\mathcal{C}})(v) = \omega_{\mathcal{C}}(X_{H_\alpha}|_{\mathcal{C}}, v) = 0.$$

Shrink to a submersion chart with connected local fibre slices. The restricted Hamiltonian is constant on each slice, so it is the pullback of a smooth local function h_α on the quotient chart. This is a local assertion: equality along connected slices does not assert equality of arbitrary integration constants on disconnected global fibres. If a function on all of S_{phys} is desired in the displayed notation, shrink the quotient neighbourhood again and use a smooth cutoff to extend this local function; the conclusion concerns only that smaller neighbourhood.

Choose any smooth local extension $\hat{h}_\alpha$ of π^*h_α . Then $H_\alpha - \hat{h}_\alpha$ vanishes on $\mathcal{C} \cap U$. The regular-ideal factorization in the proof of Theorem 7.5 gives

$$H_\alpha - \hat{h}_\alpha = \sum_a u_\alpha^a \phi_a.$$

Moreover $\omega_{\mathcal{C}} = \pi^*\omega_{\text{phys}}$ and $d(H_\alpha|_{\mathcal{C}}) = \pi^*dh_\alpha$ show that the projected vector field is X_{h_α} . AER-P(ii) identifies these reduced fields across presentations; AER-P(iii) records that their lifted differences lie in the same vertical distribution. On a connected symplectic quotient chart, functions with the same Hamiltonian vector field differ by a constant, since $X_f = 0$ implies $df = 0$.

In the trivial reduced-evolution case write $h_\alpha = c_\alpha$ on this chart. The freedom to choose the zero representative in the statement uses precisely the dynamically inert constant allowed by AER-P(ii): define

$$H_\alpha^\circ := H_\alpha - c_\alpha.$$

Choose the constant extension $\hat{h}_\alpha = c_\alpha$ in this case. The normalization leaves the Hamiltonian vector field unchanged and the regular-ideal factorization gives

$$H_\alpha^\circ = \sum_a u_\alpha^a \phi_a, \quad H_\alpha = c_\alpha + \sum_a u_\alpha^a \phi_a.$$

Thus the pure-multiplier clause is a statement about the normalized representative. Weak equality alone does not discard a nonzero constant. For a general reduced evolution the $\hat{h}_\alpha$ term remains, as in the first part of the theorem. $\square$

Compatible lifts for a chosen reduced evolution. The compatibility conditions admit a direct local construction once a reduced Hamiltonian is supplied. Work on a quotient chart for which $\omega_{\mathcal{C}} = \pi^*\omega_{\text{phys}}$ and $\mathcal{D} = \ker \omega_{\mathcal{C}}$, and choose $h \in C^\infty(S_{\text{phys}})$ on that chart. Let $\hat{h}$ be any smooth ambient extension of π^*h , and let ϕ_a be local regular defining constraints. For arbitrary smooth coefficients u^a , set

$$H_u = \hat{h} + \sum_a u^a \phi_a.$$

Each X_{H_u} is tangent to $\mathcal{C}$: the restriction $H_u|_{\mathcal{C}} = \pi^*h$ has zero differential on the characteristic directions spanned by $X_{\phi_a}|_{\mathcal{C}}$, so $d\phi_a(X_{H_u})|_{\mathcal{C}} = -dH_u(X_{\phi_a})|_{\mathcal{C}} = 0$. Moreover, $\iota_{X_{H_u}|_{\mathcal{C}}} \omega_{\mathcal{C}} = \pi^*dh$, hence $d\pi(X_{H_u}|_{\mathcal{C}}) = X_h \circ \pi$. Finally,

$$(X_{H_u} - X_{H_v})|_{\mathcal{C}} = \sum_a (u^a - v^a) X_{\phi_a}|_{\mathcal{C}} \in \mathcal{D}.$$

Thus the chosen h has a local family of compatible multiplier lifts.

Conversely, suppose a designated local family of ambient Hamiltonians preserves $\mathcal{C}$ and projects to one reduced vector field X_h . On connected local fibre slices its restricted Hamiltonians descend to functions h_α . Since the reduced quotient is symplectic, $X_{h_\alpha} = X_h$ implies $h_\alpha = h + c_\alpha$ on a connected quotient chart. Regular-ideal factorization therefore gives $H_\alpha = \widehat{h} + c_\alpha + \sum_a u_\alpha^a \phi_a$ locally; the constants can be removed only by choosing the dynamically equivalent normalized representatives. This construction and converse classify compatible lifts relative to a chosen reduced evolution. They do not select h , or establish that the Hamiltonians stipulated by a physical model form such a designated presentation family.

The bare existential form of AER-P has such local witnesses, including the choice $h = 0$. Applying it to a physical model still requires identifying the model's designated presentation family and its continuation target. Once projectability and equality of the reduced fields hold, their difference lies in $\ker d\pi = \mathcal{D}$, so clause (iii) records a consequence of (i)–(ii) explicitly. The named AER-P clauses are retained throughout the theorem and dependency ledger.

Corollary 7.9 (AER normal form). *An AER physical interior with presentation arbitrariness has, locally, the constrained-Hamiltonian normal form: coisotropic standing surface, characteristic redescription, local first-class constraints, and multiplier-type presentation Hamiltonian.*

Proof. Combine Theorems 6.3, 7.5, and 7.8. □

Theorem 7.10 (Local AER forcing theorem, in-paper form). *The local constrained-Hamiltonian structure used by the main gravity theorem is proved within this manuscript rather than imported as a black box. Specifically, under the regular-stratum AER interface, quotient structure yields an involutive redescription distribution, observable Poisson descent and separation force characteristic capture and coisotropic standing, AER-Max eliminates hidden presymplectic remainders, coisotropic normal form supplies local constraint generators, first-class closure follows from regularity, and AER-P yields multiplier-type Hamiltonian presentation.*

Proof. The quotient submersion gives $\mathcal{D} = \ker(d\pi)$ and involutivity by Lemma 5.7. AER-Obs_{min} and AER-Sep force $(TC)^\perp \subseteq \mathcal{D}$ and hence coisotropicity by Lemma 6.1. AER-Max gives $\mathcal{D} \subseteq \ker(\omega_{\mathcal{C}})$ by Lemma 6.2. Theorem 6.3 combines the two inclusions to identify admissible redescription with the characteristic foliation. The local coisotropic normal form then gives constraint generators in Definition 7.2; Theorem 7.5 proves weak closure of their Poisson brackets. Finally, Theorem 7.8 proves the reduced-plus-multiplier decomposition of presentation Hamiltonians under AER-P. These steps establish the local theorem object used in Theorem 8.2. □

8 The AASC gravity theorem

We now identify the structure derived above with gravity in the AASC sense of Definition 2.1. This section is the conceptual center of the paper: it argues that gravity is not merely a name for curvature, but the necessary constraint role of admissible physical interiority.

Definition 8.1 (GR-type canonical constraint architecture). *A physical theorem object has GR-type canonical constraint architecture when its standing-bearing interior contains, up to lawful role-preserving equivalence:*

- (i) *a standing surface $\mathcal{C}$ in a Hamiltonian state carrier;*
- (ii) *physical observables factoring through a quotient $\pi : \mathcal{C} \rightarrow S_{\text{phys}}$;*
- (iii) *admissible redescription coinciding with the characteristic foliation of $\omega_{\mathcal{C}}$;*

- (iv) *local first-class constraint generators spanning admissible redescription;*
- (v) *presentation-parametrized dynamics in multiplier form, with representative evolution differing by redescription directions.*

The definition is structural. It does not require the metric variables, Einstein equations, or ADM presentation to be primitive.

The theorem classifies structure at the level of constraint architecture; dynamical specialization, such as Einstein equations versus modified gravity theories, lies within this structure and is not fixed by the present result.

The Hamiltonian form arises under the AER specialization; equivalently, any admissible formulation must realize an isomorphic constraint/continuation structure (Section 5) to remain admissible.

Theorem 8.2 (Gravity Necessity Theorem — AASC form). *Let a system instantiate an admissible physical interior in the regular-stratum AER comparison class and satisfy presentation arbitrariness AER-P. Then it necessarily instantiates gravity in the AASC sense: a GR-type canonical constraint architecture. Therefore, within that comparison class, a world lacking the AASC gravity role is not an admissible non-gravitational physical interior; it fails to instantiate the target.*

The theorem does not replace empirical gravitational theory; it identifies the structural condition under which such theory is possible.

Proof. By Theorem 3.7, an admissible physical interior must instantiate admissibility, standing, reference, and irreversibility. By Theorem 4.4, reusable physical standing under lawful redescription factors through quotient structure. The theorem’s regular-stratum AER hypothesis supplies the particular Hamiltonian interface of Definition 5.12: a standing surface, quotient projection, physical observables, redescription distribution, and the three declared AER compatibility clauses. The kernel and quotient results fix their admissibility roles; the stated specialization fixes the smooth and observable data on which the local forcing argument operates.

AER-Obs_{min}, AER-Sep, and AER-Max then force $\mathcal{D} = \ker(\omega_C)$ by Theorem 6.3. Thus the standing surface is coisotropic and admissible redescription is the characteristic foliation. The coisotropic local normal form yields local constraint generators; Theorem 7.5 proves their first-class closure. The assumed presentation arbitrariness AER-P then gives the reduced-plus-multiplier Hamiltonian presentation by Theorem 7.8, including the common reduced dynamics and vertical representative freedom. These are exactly the clauses of Definition 8.1, and by Definition 2.1 this is gravity in the AASC sense.

Suppose now that the same target lacked the AASC gravity role. Then at least one clause in Definition 8.1 fails. If the standing surface or quotient fails, reusable physical identity fails. If characteristic redescription fails, quotient identity and lawful redescription no longer agree. If first-class closure fails, admissible continuation is not closed on the standing surface. If multiplier-type presentation fails in the presence of AER-P, presentation freedom carries untyped physical load. Each case violates the admissible physical interior rather than producing a same-scope non-gravitational realization. Hence the AASC gravity role is forced. $\square$

Corollary 8.3 (Non-contingency of gravity in the fixed target). *Within the AER comparison class, gravity in the AASC sense is not a contingent sector added to matter or quantum structure. It is a necessary structural condition for the admissible physical interior.*

Proof. Immediate from Theorem 8.2. Removing the GR-type canonical constraint architecture removes the conditions under which the interior can be admissibly instantiated; by Definition 2.1, that architecture is precisely the AASC gravity role. $\square$

Remark 8.4 (What “forced gravity” does and does not mean). The phrase “forced gravity” is comparison-class relative and role-theoretic. The theorem does not say that every conceivable formal system contains Lorentzian geometry, nor that the Einstein equations are derived with all empirical coefficients. It says that any system claiming to instantiate the same admissible physical interior, with Hamiltonian observable evaluation, quotient completeness, and admissible continuation, must instantiate the constraint role identified in Definition 2.1. Canonical GR is the standard realization of that role.

Proposition 8.5 (Curvature as gravitational projectional bookkeeping). *In the AASC reading of gravity, curvature is not denied; it is typed as a geometric presentation of the standing-bearing constraint architecture that constitutes gravity in the AASC sense. Its primitive status is not required for the proof of Theorem 8.2.*

Proof. Theorem 8.2 derives the AASC gravity role from quotient, observable, and continuation constraints, not from a primitive metric. A metric or Lorentzian geometric presentation may faithfully realize the resulting tensor content, but the proof does not require it as a primitive substrate. By Theorem 4.7, geometry is anchor, tensor, or skin relative to a fixed target: boundary/domain choices are anchors, curvature invariants and observable gravitational data are tensors, and coordinate or gauge presentation is skin. Therefore curvature remains physically meaningful as projected tensor content while not serving as the deepest same-domain foundation. $\square$

Extent of the role classification. An ordinary constrained Hamiltonian model can instantiate the role if it satisfies the entire declared interface. Its inclusion is consistent with a role-theoretic definition whose realization need not yet specify a metric, field content, or numerical interaction law. The mathematical content is the forcing of the listed geometric and Hamiltonian properties from that interface; the identification of this role with AASC gravity is fixed by Definitions 2.1 and 8.1.

The regular case $\mathcal{C} = S$, $\mathcal{D} = 0$ is also allowed. Here coisotropy is automatic, the constraint list is empty, and the Hamiltonian formula reduces to its reduced term. If that reduced evolution is trivial, the normalized representative is zero. This limiting case does not assert a nonzero gravitational interaction. Neither $k > 0$ nor $\mathcal{D} \neq 0$ is needed for the stated role theorem. Inclusion of this formal limit does not automatically admit a named Galilean effective regime: the scope guard in Remark 5.2 and the full AER interface still apply.

Necessity and native realization. Theorem 8.2 establishes the necessity of the AASC gravity role throughout its stated comparison class, including $k = 0$. The coisotropic, characteristic, first-class, and multiplier conclusions follow from the forcing argument; the role definition identifies the architecture whose necessity is proved. A same-scope admissible interior cannot dispense with this architecture. Canonical GR realizes it with the GR constraint manifold as standing surface. Metric projections, Einstein dynamics, and specified couplings belong to the native-realization results for their declared targets [22]; those target-specific premises leave the structural necessity result intact.

Intuitive compression. Every physical interior satisfying the stated AER hypotheses and AER-P must instantiate gravity in the AASC sense. Theorem 8.2 forces its constraint architecture; absence of that role is failure to instantiate the same target. This necessity includes the zero-constraint endpoint. Canonical GR supplies the standard metric realization.

9 Canonical GR as a downstream realization

The previous sections show that GR-type structure is forced. Canonical GR supplies the familiar geometric realization of that structure. This section records the standard ADM chain [5, 4] only to exhibit the match; it is not used as a premise in the forcing proof.

The displayed formulae use vacuum bulk ADM conventions with the overall gravitational coupling absorbed into the units. Smearing functions are taken on a closed spatial slice or with support and boundary conditions that remove surface contributions. Where physical boundary charges survive, the differentiable Hamiltonian includes the corresponding boundary terms; their standing-bearing content belongs to the declared boundary and continuation data. The finite-dimensional local theorem does not itself construct an infinite-dimensional ADM quotient. The field realization requires its own regularity, differentiability, constraint and quotient control on the chosen sector.

Obligations for a field realization. A field application must establish the analogues it uses in its own functional setting. The following ledger identifies where the finite-dimensional argument needs field-specific justification; it is not an infinite-dimensional extension theorem.

Part of the argument	Field-specific obligation
Carrier and duality	Specify the function spaces, topology and differentiability class; justify the symplectic and conormal operations actually used. Weak nondegeneracy alone does not supply a Darboux chart.
Standing and generators	Establish regularity of the chosen constraint sector and the generating or factorization statement needed there. Account for reducibility and any failure of a global constraint basis.
Quotient and observables	Establish the local quotient domains and the stated observable algebra, including extension independence, separation and maximality where invoked. A global quotient or groupoid description needs its own argument.
Brackets and boundaries	Define differentiable smeared generators, their domains, boundary conditions and charges; verify closure with the chosen bracket.
Continuation	Establish the required local flows, their standing preservation and projectability, and the designated common reduced evolution.

The distinction between strong and weak symplectic forms matters: Darboux’s theorem can fail for a weak symplectic form even on a Hilbert space [21]. Naming a Banach or Fréchet setting therefore does not discharge the normal-form step. Likewise, a successful local quotient argument neither supplies global Hausdorffness nor classifies residual global identifications as skin. Later field

realizations may discharge their own obligations without becoming premises of the local gravity-necessity proof.

Let $(M, g_{\mu\nu})$ be globally hyperbolic and foliated by spacelike hypersurfaces Σ_t . The ADM line element is

$$ds^2 = -N^2 dt^2 + h_{ij}(dx^i + N^i dt)(dx^j + N^j dt),$$

where N is the lapse and N^i the shift. The extrinsic curvature is

$$K_{ij} = \frac{1}{2N} \left(\dot{h}_{ij} - D_i N_j - D_j N_i \right).$$

The Einstein-Hilbert action with the boundary term decomposes, up to total derivatives, into the ADM Lagrangian density

$$\mathcal{L}_{ADM} = N\sqrt{h} \left(R(h) + K_{ij}K^{ij} - K^2 \right).$$

The canonical momentum conjugate to h_{ij} is

$$\pi^{ij} = \sqrt{h} (K^{ij} - h^{ij} K).$$

No time derivatives of N or N^i occur; they are multipliers. The canonical Hamiltonian density is

$$\mathcal{H}_{can} = N\mathcal{H} + N^i \mathcal{H}_i,$$

where

$$\mathcal{H} = \frac{1}{\sqrt{h}} \left(\pi^{ij} \pi_{ij} - \frac{1}{2} \pi^2 \right) - \sqrt{h} R(h), \quad \mathcal{H}_i = -2D_j \pi^j_i.$$

The constraint surface is

$$\mathcal{C}_{ADM} = \{(h, \pi) : \mathcal{H}(x) = 0, \mathcal{H}_i(x) = 0 \forall x \in \Sigma\}.$$

With smeared constraints

$$H[N] = \int_{\Sigma} N \mathcal{H} d^3x, \quad H[\vec{N}] = \int_{\Sigma} N^i \mathcal{H}_i d^3x,$$

the hypersurface deformation algebra is

$$\begin{aligned} \{H[\vec{N}], H[\vec{M}]\} &= H[\mathcal{L}_{\vec{N}} \vec{M}], \\ \{H[\vec{N}], H[M]\} &= H[\mathcal{L}_{\vec{N}} M], \\ \{H[N], H[M]\} &= H_i[h^{ij}(N\partial_j M - M\partial_j N)]. \end{aligned}$$

This is first-class closure with structure functions.

The following standard ADM comparison is read on a sector where the field-specific operations and identifications just listed are justified. It identifies the canonical structural realization; it does not establish those analytic conditions from the finite-dimensional theorem.

Proposition 9.1 (ADM realizes the AER normal form). *On its regular constraint surface, canonical GR realizes the AER normal form: coisotropic standing, first-class redescription, and multiplier-type Hamiltonian presentation.*

Proof. The ADM constraint surface is the standing surface. The constraint flows generate spatial diffeomorphism and refoliation redescription on the surface. The hypersurface deformation algebra gives first-class closure. The total Hamiltonian is a multiplier combination of constraints with lapse and shift as presentation parameters. Thus the ADM realization satisfies the clauses of Definition 8.1. $\square$

Remark 9.2 (Direction of dependence). The ADM chain is not the source of the AASC theorem. Rather, once the AER rigidity theorem has been proved, ADM is recognized as a concrete geometric presentation of the forced structure.

10 Fixed-domain closure of the GR-type interior

The forcing theorem shows that GR-type structure appears once the AER interface is fixed. A further question remains: could a non-equivalent same-scope rival preserve the target while carrying additional gravitational standing? This section closes that route.

Definition 10.1 (Derived AER gravitational object). *The derived AER gravitational object is*

$$\mathcal{G}_{AER} := (\mathcal{C}, \pi : \mathcal{C} \rightarrow S_{\text{phys}}, \mathcal{D} = \ker(\omega_{\mathcal{C}}), \{\cdot, \cdot\}_{\text{phys}}, h_{\text{phys}})$$

where h_{phys} is included when AER-P supplies a reduced generator. Local constraint bases and multiplier coefficients are presentations, not additional same-scope structure.

Definition 10.2 (Same-scope gravitational presentation). *A same-scope presentation of $\mathcal{G}_{AER}$ preserves, up to lawful reparameterization: (i) $\mathcal{C}$ and π ; (ii) $\mathcal{D} = \ker(\omega_{\mathcal{C}})$; (iii) the physical observable algebra and its descended bracket; and (iv) the induced physical continuation on S_{phys} when present.*

What remains fixed. The comparison in this section holds $\mathcal{C}$, π , the characteristic distribution, the observable algebra and bracket, and the physical continuation fixed as in Definition 10.2. Changing the reduced Hamiltonian so that its vector field changes therefore changes this theorem object. The closure result concerns presentations of the specified object; it does not select one dynamical law from all laws that can share an abstract constraint pattern. In particular, discrete quotient data remain part of π and are not discarded merely because they are invisible in $\mathcal{D}$.

Theorem 10.3 (Two-locus bridge). *Once the AER normal form is derived, same-scope gravitational standing-bearing content is exhausted by two loci:*

- (i) *quotient physical content on S_{phys} , including physical observables, their bracket, and induced physical continuation;*
- (ii) *characteristic/redescription structure on $\mathcal{C}$, equivalently $\mathcal{D} = \ker(\omega_{\mathcal{C}})$ and its local first-class span.*

If two same-scope presentations agree on these loci, they agree on physical observables, admissible redescription, and induced physical continuation. No third same-scope gravitational locus survives.

Proof. By Theorem 6.3, admissible redescription equals the characteristic foliation. By Theorem 7.5, local first-class generators present that foliation. By Theorem 7.8, presentation-dependent Hamiltonians differ from reduced content only by multiplier combinations of the same constraints. Hence the only standing-bearing differences left are changes to quotient physical content or changes to characteristic/redescription structure. If neither changes, all remaining differences are coordinate choices, representative extensions, constraint bases, multipliers, or gauge descriptions, hence skin by Theorem 4.7. $\square$

Definition 10.4 (Same-scope rival). *A genuine same-scope rival to $\mathcal{G}_{AER}$ is a presentation that claims to preserve Definition 10.2 while adding physically relevant same-domain structure not lawfully equivalent to the derived object.*

Definition 10.5 (Normal-form family for gravitational rivals). *After quotienting out skin, every alleged same-scope gravitational rival has one of the following first divergence types:*

- (N1) *anchor motion: a precondition, boundary, regularity class, or admissibility envelope changes;*
- (N2) *quotient-locus motion: physical content on S_{phys} changes;*
- (N3) *characteristic-locus motion: redescription/constraint structure changes;*
- (N4) *bridge-neutral insertion: an auxiliary distinction is introduced while changing neither bridge locus;*
- (N5) *selector/repair route: a bridge-neutral distinction is later made physical by a selector, deferred activation, post-selection, or same-theory reset.*

Theorem 10.6 (Nonempty normal-form image). *Every genuine same-scope rival has a nonempty image in the finite family of Definition 10.5.*

Proof. Choose the first point at which the rival differs physically from $\mathcal{G}_{AER}$ after quotienting skin. If a precondition for the theorem object changes, this is anchor motion. Otherwise anchors are fixed. By Theorem 10.3, a physical difference must change quotient content or characteristic/redescription structure, unless it changes neither bridge locus at the point of introduction. Those cases are quotient-locus motion, characteristic-locus motion, and bridge-neutral insertion. If a bridge-neutral insertion is later claimed to carry physical force, that force must arise through deferred activation, reset, selector, or repair, giving the selector/repair route. These alternatives exhaust the first divergence. $\square$

Theorem 10.7 (Collapse of same-scope gravitational rivals). *No genuine same-scope rival to $\mathcal{G}_{AER}$ survives in the fixed AER comparison class.*

Proof. By Theorem 10.6, any rival has a nonempty normal-form image. Anchor motion changes the theorem object and is not same-scope. Quotient-locus motion changes physical observables, bracket, or induced continuation, so it changes the theorem object. Characteristic-locus motion changes admissible redescription or constraint structure, so it also changes the theorem object. A bridge-neutral insertion has no same-scope physical effect by Theorem 10.3; it is skin. If such an insertion later acquires physical force, that force is supplied by a second classifier, selector, deferred activation, post-selection, or reset. Corollary 4.8 and Lemma 3.6 exclude those as same-domain standing-preserving moves. Therefore every alleged rival is skin, scope change, declared new load, or standing failure. $\square$

Corollary 10.8 (Uniqueness of the GR-type same-scope interior). *Within the fixed AER comparison class, the GR-type constraint architecture is the unique same-scope admissible realization of the observable-compatible quotient structure.*

Proof. Theorem 8.2 gives existence of GR-type structure under the AER conditions. Theorem 10.7 excludes non-equivalent same-scope rivals. $\square$

11 Projection architecture and primitive quantization

Gravity has already been forced as the constraint role of Definition 2.1. This section treats the further problem of realizing gravitational and quantum outputs in one physical target. Three claims have different proof burdens: common anchor/tensor/skin governance, actual native projections from a common carrier, and a direct map from one projected output to another. The first is fixed by the kernel for every determinate joint target. The second requires a joint realization and verified native maps. The third requires an additional physical-fibre condition. None supplies a metric or quantum premise to Theorem 8.2.

Definition 11.1 (Architectural projection map). *Let*

$$\mathcal{A} = (\text{Anc}_{\mathcal{A}}, \text{Ten}_{\mathcal{A}}, \text{Skin}_{\mathcal{A}}, \sim_{\mathcal{A}}, \text{Cont}_{\mathcal{A}})$$

be an admissibility-bearing quotient architecture; also write $\mathcal{A}$ for its presentation carrier. Let $q_{\mathcal{A}} : \mathcal{A} \rightarrow \mathcal{A}_{\text{phys}}$ and $q_X : X \rightarrow X_{\text{phys}}$ be the lawful physical quotients. An architectural projection to the independently declared theorem object X is a map $\Pi_X : \mathcal{A} \rightarrow X$ inducing a surjective physical map $\bar{\Pi}_X : \mathcal{A}_{\text{phys}} \rightarrow X_{\text{phys}}$, with these properties:

- (P1) **Anchor preservation:** *the map preserves the domain-fixing anchors required by the X -role. Every admissible anchor variation of the full declared X target has a lawful lift in $\mathcal{A}$, with its prescribed domains and endpoints.*
- (P2) **Tensor preservation:** *the independently fixed standing-bearing witnesses of X are realized by the corresponding tensor witnesses in $\mathcal{A}$. Their native interpretation and domains agree, and lawful skin creates no standing-bearing witness.*
- (P3) **No loss of role-typed standing:** *fix independently of Π_X a set-indexed family U_X of all required X -typed standing uses on $\mathcal{A}_{\text{phys}}$, including their domains and admissible contexts. Write E_X for contextual agreement on their definedness and values. Require*

$$\text{Fib}(\bar{\Pi}_X) \subseteq E_X, \quad \text{Fib}(F) = \{(a, a') : F(a) = F(a')\}.$$

Thus replacing any argument of a partial, possibly multi-input use by another with the same physical image preserves whether the use is defined and its value when defined.

- (P4) **Quotient preservation:** *lawful equivalence in $\mathcal{A}$ maps to lawful equivalence in X , so $\bar{\Pi}_X([a]) = [\Pi_X(a)]$. All claimed target uses have this physical identity discipline; any additional smooth or analytic structure requires its own verification.*
- (P5) **Continuation preservation:** *claimed source histories project to lawful target histories with compatible domains, endpoints, identities and composition, and every admissible target history required by the full declared target is realized by a source history up to lawful equivalence. For partial deterministic histories, projectability requires fibre-saturated domains and fibre-congruent outputs. Relational histories require the corresponding relational compatibility checks.*

Surjectivity concerns the full declared physical target, not every redundant raw representative. Replacing that target by a proper realized image is a stated scope restriction.

The quantifiers in P1 and P5 range over the independently declared target instance and its admitted variations and histories. For a frozen certificate, changing a fixed source, potential, distinguished initial datum, seed or matching package specifies another target instance. Clock, domain and failure information required for the declared histories remains part of their context. This reading preserves the full target rather than enlarging it to unrelated parameter choices [28, Definition 23.2; Lemma 46.1; Theorem 70.5].

Role uses, physical fibres, and lawful skin. The set-indexed family U_X is fixed from the joint presentation's role ledger before testing a proposed map; its native counterpart is independently fixed and matched in P2. Defining it after the fact as precisely the uses visible through that map would make P3 vacuous. A full-target claim requires this ledger to cover every standing use required by the independently declared target, with history coverage checked separately under P5; omission of a required witness or history leaves the original full-target claim uncertified. An explicitly narrower target must be assessed separately. A distinction invisible to one role can remain tensor content for another. Equality of an individual projection's outputs therefore does not establish global skin. Even equality of a joint probe must be compared with the lawful presentation groupoid: a claim of maximal joint skin requires both preservation by lawful skin arrows and a proof that the proposed identifications are lawful. The pair of role outputs need not detect every joint incidence.

Physical-fibre descent. Let $F : A \rightarrow B$ and let $u : D_u \subseteq A^n \rightarrow V_u$ be an independently fixed partial use. Suppose substitution within an F -fibre in any argument preserves definedness and value in every context. Define

$$D_{\tilde{u}} = F^n(D_u), \quad \tilde{u}(F(a_1), \dots, F(a_n)) = u(a_1, \dots, a_n).$$

Replacing representatives one argument at a time proves that this is well defined and that $D_u = (F^n)^{-1}(D_{\tilde{u}})$. Surjectivity gives uniqueness. This proves the descent needed in P3; the reverse inclusion $E_X \subseteq \text{Fib}(\bar{\Pi}_X)$ is an additional exact role-completeness assertion. For a partial state evolution $c : D_c \subseteq A \rightarrow A$, one must separately show saturation of D_c and

$$F(a) = F(a') \implies F(c(a)) = F(c(a')) \quad (a, a' \in D_c).$$

Then $c_B(F(a)) = F(c(a))$ is well defined. Covering the independently declared target history class remains a further P5 requirement.

Definition 11.2 (Projection equivalence). *Two theorem objects X and Y are projection-equivalent for a declared joint target when one admissibility-bearing architecture realizes that target and admits architectural projections to both objects. Each projection covers its declared physical target and satisfies Definition 11.1. The claimed cross-role witnesses and continuation incidences are independently specified in the same joint target, invariant under lawful equivalence, and verified against the paired outputs wherever claimed. Individual role completeness, completeness of a joint probe and maximal lawful skin are separate assertions requiring separate proofs.*

Definition 11.3 (Gravitational and quantum projection targets). *The gravitational target G has as its standing-bearing content the GR-type canonical constraint architecture of Definition 8.1. The quantum target Q comprises an independently established Hilbert/Schrödinger predictive interior together with the declared boundary-witness and measurement-record quotient structure. Their regimes, anchors, witnesses, identities and continuation classes are fixed before testing a joint realization. Specifying these separate targets does not assert the existence of a common carrier or native projections realizing both.*

Native targets and their scope. The constraint role established here precedes a Lorentzian metric realization. The Einstein dynamics theorem classifies a faithful metric projection under its stated metric, response and source conditions; those are further native-target data [22]. The Schrödinger dynamics theorem uses a standing quotient and a separately constructed faithful spatial projection with its Hamiltonian, domain and scale conditions [23]. Its closed predictive result does not by itself supply all boundary-record structure in Q . Likewise, the occupied WDW physical line and a spatial projective-Hilbert target have distinct carrier types. The earlier physical-quotient source distinguishes predictive and record roles and proves common role classification [24]; that classification does not itself construct the native joint realization required here.

Role	Gravitational target	Quantum target
Anchors	regular stratum, standing surface, boundary and scope data	predictive regime, Bell scope, record domain
Tensor content	quotient observables, characteristic foliation, first-class span, multiplier continuation	Hilbert geometry, unitary continuation, boundary witnesses, record quotient
Lawful skin	equivalent coordinates, gauges, constraint bases and extensions in the fixed target	equivalent bases and representatives; phase only where physically inert in the declared target

Joint realization established in the later corpus

The structural-unification result concerns one determinate physical target bearing metric deformation-response standing and quantum transition/record standing. Its no-split and same-interior results place the faithful paired projections in one admissible interior; its main theorem gives the paired first-response dynamics in their declared native regimes [26, Theorems 5.1, 6.5 and 48.1].

The constructive Modules A–D chain supplies a nonempty common realization and completes its certificate: Module C constructs the upstream carrier and four role processes, Module D completes the tensor and joint-skin interfaces, and the capstone proves compatible occupation and complete realization of its exact frozen target [25, 27, 28]. The local theorem below gives the general comparison criterion. The subsequent fixed-target correspondence verifies it for the occupied closed-packet instance using the already constructed objects and their declared history scope.

Theorem 11.4 (Canonical role profiles and native projection equivalence). *Let $P \neq \emptyset$ be an independently supplied jointly instantiated physical presentation, with lawful identity $\approx_P$, the admissibility-bearing role data of Definition 4.5, joint incidences and admissible histories. Its standing uses and their domains are invariant under $\approx_P$. Its partial deterministic histories have $\approx_P$ -saturated domains and $\approx_P$ -congruent outputs, and therefore induce histories on the physical quotient; identities and composition respect this equivalence. Put $A = P/\approx_P$. For each $X \in \{G, Q\}$, fix from this presentation's role ledger, before proposing any native map, a set-indexed family of the required partial role uses U_X on A ; let E_X be contextual agreement on their definedness and values.*

- (i) *The canonical role profiles $R_X = A/E_X$ and surjections $e_X : A \twoheadrightarrow R_X$ exist as sets. All uses in U_X descend uniquely. The induced A -histories with E_X -saturated domains and E_X -congruent outputs descend as well, preserving the source identity and composition laws wherever these are specified.*

(ii) Suppose, additionally, that the independently established native targets G, Q admit supplied raw maps $f_X : P \rightarrow X$, with P their common declared domain of eligibility. Require $F_X = q_X \circ f_X$ to be constant on $\approx_P$ -classes, surjective onto the full declared X_{phys} , and to induce $\bar{f}_X : A \rightarrow X_{\text{phys}}$ with $\text{Fib}(\bar{f}_X) \subseteq E_X$. Verify matching of native anchor values and domains and lawful lifts of every admissible anchor variation; native witness/domain matching and invariance under lawful skin; and the full P5 conditions, including saturated domains, congruent outputs and realization of every admissible history of the full declared native target with compatible identities and composition. Verify also the declared cross-role incidences. Then these maps realize projection equivalence of G and Q for this joint target. There is a unique surjection $h_X : X_{\text{phys}} \rightarrow R_X$ such that $e_X = h_X \circ \bar{f}_X$; it is a bijection exactly when $\text{Fib}(\bar{f}_X) = E_X$.

The construction of role profiles in (i) does not identify them with native metric or Hilbert targets. Part (ii) states the additional realization obligations explicitly; it does not derive the supplied joint carrier and native maps from the existence of the two separate targets. Analytic regularity is a further obligation in its declared category.

Proof. Lawful invariance first makes each specified use well defined on A . Contextual agreement is an equivalence relation: reflexivity and symmetry are immediate, and transitivity follows by two successive substitutions in each fixed context. Thus R_X exists as a set quotient, with $\text{Fib}(e_X) = E_X$. The physical-fibre descent argument gives the unique partial uses on R_X , including their domains. The separate history congruence hypothesis defines each descended history by $[a]_{E_X} \mapsto [c(a)]_{E_X}$. Saturation gives its domain, and applying the quotient map to a source identity or composite proves the corresponding law on the image. This proves (i).

For (ii), define $\bar{f}_X([p]) = F_X(p)$. Constancy on $\approx_P$ -classes proves well-definedness and P4, and physical coverage proves surjectivity. The asserted fibre inclusion is precisely P3, so the independent role uses descend through $\bar{f}_X$. Define

$$h_X(\bar{f}_X(a)) = e_X(a).$$

The inclusion makes h_X well defined; surjectivity of $\bar{f}_X$ gives uniqueness, and that of e_X gives coverage of R_X . It is injective exactly when $aE_X a'$ also implies $\bar{f}_X(a) = \bar{f}_X(a')$. This is the additional equality of fibre relations.

Equip the original carrier P with its supplied anchor, tensor, lawful-skin and history data as $\mathcal{A}$, retaining its actual joint incidences on A . The native anchor, witness and continuation verifications are the additional inputs P1, P2 and P5; they are not consequences of the set quotient. Together with the derived physical maps and the checked cross-role incidences, they establish Definition 11.2. No step replaces the actual common carrier by the product $R_G \times R_Q$, which could introduce unrealized pairings or discard joint incidence information. Nor does it identify E_G, E_Q , or their intersection with lawful global skin without a separate joint-completeness proof. $\square$

Fixed-target correspondence with the completed closed packet. Use the occupied closed packet of Module C, Definitions 24.1–24.5, imported by capstone Theorem 37.3, with its accepted GR/source, quantum, boundary and matching occupants (Definition 45.1; Theorems 47.4 and 70.5). Write $\mathfrak{G}_*$ for that fixed AER constraint object, m_0 for its vacuum metric/source orbit, $\mathcal{Q}_*$ for the full projective carrier with its fixed oscillator process R_t , $\mathcal{R}_{\text{MR},*}$ for the record quotient of the declared record use, and c_0 for its fixed factorized compatibility class. These are the independently specified native ingredients of this packet. Retain its certified clock interval J and failure boundary as history context. The ambient fixed-clock presentation carrier, physical native endpoints and

maps are

$$\begin{aligned}\hat{P}_* &= J \times \mathcal{Q}_* \times \mathcal{R}_*, & \hat{G}_*^\tau &= J \times \{(\mathfrak{G}_*, m_0)\}, \\ \hat{Q}_*^\tau &= J \times \mathcal{Q}_* \times \mathcal{R}_{\text{MR},*} \times \{c_0\}, \\ F_G(\theta, q, r) &= (\theta, \mathfrak{G}_*, m_0), \\ F_Q(\theta, q, r) &= (\theta, R_\theta q, [r]_{\text{MR}}, c_0).\end{aligned}$$

The operator, source, domain and boundary certificates remain fixed structure on these carriers. The displayed product is justified by this packet's independently varying ray and retained-record coordinates and fixed c_0 ; general joint targets retain their incidence conditions. Let P_*, G_*^τ, Q_*^τ carry the certificate's predeclared initial-data and eligible-orbit restrictions, and restrict the displayed maps accordingly. With no pointed restriction they are the displayed carriers. A fixed initial object specifies its permitted orbit tube and subhistories, with the same matched native initial data; it does not freeze the evolving clock or record at their initial values. Permitted local durations satisfy $\theta + [0, t] \subset J$. The clock coordinate retains existing history context and supplies no premise to gravity necessity.

Verification. Every admitted $(\theta, y, [s]_{\text{MR}}, c_0)$ has the explicit lift $(\theta, R_{-\theta}y, s)$, choosing a record representative consistent with any fixed initial data; the gravitational output is covered at each admitted clock value. The constructed upstream flow

$$\alpha_t(\theta, q, r) = (\theta + t, q, r + t)$$

projects to the fixed vacuum development and to $(\theta + t, R_t y, [s + t]_{\text{MR}}, c_0)$. Equal role outputs retain the same θ , so their permitted local domains agree. Module C's fibre preservation then gives congruent outputs. Choosing one initial lift produces every declared target segment and its compositions. For fixed initial data, the native initial state is matched by the certificate, and the displayed intertwining carries its entire permitted orbit to the native orbit; thus the restricted carriers and histories are covered as well. This verifies physical coverage and P5, including domain saturation. Retaining the clock context is essential: untagged equal rays can have different distances to the boundary of J . The independent complete role profiles, lawful quotient functors, fixed anchors and capstone incidence certificate supply P1–P4 and the joint conditions, as recorded in Appendix A. Hence Theorem 11.4(ii) applies to these structured native endpoints in the completed closed-packet scope. The gravitational endpoint retains $\mathfrak{G}_*$ as well as its metric output; the quantum endpoint retains the record and compatibility roles. The full Stone group classifies the quantum process, while the admitted joint histories retain their local clock and failure scope.

Remark 11.5 (Projection is not formal equivalence). Projection equivalence is architectural realization for the declared joint target. It does not identify ADM phase space, Lorentzian geometry, Hilbert space, operator algebras and records as one mathematical formalism. The canonical role-profile construction is an in-manuscript quotient result; native-target realization in Theorem 11.4(ii) is explicitly conditional on its supplied maps and verified obligations. The general gravity-necessity theorem remains independent of those inputs.

Definition 11.6 (Primitive same-domain quantization claim). *For a fixed jointly realized target, a primitive quantization claim asserts a same-domain standing-bearing map from its gravitational theorem object to its quantum theorem object, realizing the fixed pairing while carrying independent foundational authority beyond the governing architecture. At physical level, compatibility with that pairing means $\bar{\Pi}_Q = q \circ \bar{\Pi}_G$. A raw representative map must also pass the quotient and native-structure checks; a physical set map alone does not supply it.*

Theorem 11.7 (Projection-to-projection obstruction). *Fix a jointly realized target with architectural projections to G, Q . They determine the span*

$$G_{\text{phys}} \xleftarrow{\bar{\Pi}_G} \mathcal{A}_{\text{phys}} \xrightarrow{\bar{\Pi}_Q} Q_{\text{phys}}.$$

A compatible set map q on the realized gravitational image exists if and only if

$$\text{Fib}(\bar{\Pi}_G) \subseteq \text{Fib}(\bar{\Pi}_Q).$$

When it exists it is unique there and satisfies $\bar{\Pi}_Q = q \circ \bar{\Pi}_G$. Its native physical structure requires the corresponding anchor, witness and continuation checks. In this fixed target a primitive same-domain map carrying an undeclared independent authorizing role has no standing. A proposal changing the pairing, anchors, witnesses, identity or continuation must declare the changed scope or load; a hidden change fails standing.

Proof. The pair of projections defines the relation

$$\mathcal{R}_{GQ} = \{(\bar{\Pi}_G(a), \bar{\Pi}_Q(a)) : a \in \mathcal{A}_{\text{phys}}\}.$$

If q realizes this relation, equality of gravitational images gives equality of quantum images, proving necessity of the fibre inclusion. Conversely, under the inclusion set $q(\bar{\Pi}_G(a)) = \bar{\Pi}_Q(a)$. This is independent of the representative and is defined on the entire realized gravitational image. Every compatible map must take these values, which proves uniqueness. For architectural projections that image is the full declared G_{phys} ; no values on an enlarged target are fixed by this argument.

This factorization concerns values relative to the supplied pair. The authority conclusion uses a separate fact: the determinate target is kernel-governed, and Corollary 4.8 excludes a second undeclared same-scope authorizer. With anchors and invariant standing content fixed, a compatible map cannot acquire such authority by changing its name or representation. A map with different physical outputs changes the fixed pairing or standing-bearing content. An anchor change is scope motion; new witnesses are declared or hidden load; identity or history changes are quotient or continuation motion. If hidden, the change violates the fixed-target classification and the irreversibility discipline. These statements do not assert that every bare mathematical map $G \rightarrow Q$ is nonexistent. $\square$

A common source need not give a direct map. Take the form of Module C's local packet with outputs

$$\Pi_g(\theta, q, r) = m_0, \quad \Pi_H(\theta, q, r) = R_\theta q,$$

where q is a physical ray and R_θ is an invertible ray evolution. At fixed θ, r , distinct rays have the same metric output and different quantum outputs. Hence the fibre inclusion in Theorem 11.7 fails for these two outputs, although both have the supplied common source. The packet's metric orbit is one fixed target; this observation does not identify that orbit with the full canonical gravitational target of Definition 11.3.

Corollary 11.8 (Why familiar quantization divergences appear). *Primitive gravitational quantization attempts must encounter standing-bearing divergence in representation choice, observable assignment, constraint closure, physical state selection, inner product or measure, boundary extraction, continuation, or output/record extraction, unless those choices are typed as skin or explicit new load.*

Proof. Interpret “primitive” in the fixed-target sense of Definition 11.6. A quantization tuple can affect that target through its state carrier, observable representation, constraint implementation, physical domain and inner product, measure or regulator data, boundary data, continuation law, or output extraction. A choice with no standing-bearing effect is skin; a declared effect changes anchors or tensor/continuation data; an undeclared effect is hidden load or standing failure. This is the locus-specific governance conclusion of Theorem 11.7. It predicts no particular numerical or perturbative divergence. $\square$

Remark 11.9 (Effective practice is not challenged). Perturbative quantum gravity, semiclassical gravity, gravitational EFT, loop/spinfoam methods, string-like UV completions and path-integral methods remain assessable in their declared scopes. Explicit error discipline, new anchors, tensor witnesses or realization certificates identify restricted-scope or richer-scope constructions. The obstruction concerns an undeclared independent foundational authorizer for the same fixed target.

12 Positive program: constraint-native gravity

The constructive target is a constraint-native realization with verified GR-type, QM-type and boundary projections. Theorem 11.4 separates the canonical role profiles from their native realization; the certificate records the additional physical obligations.

Definition 12.1 (Constraint-native realization). *A constraint-native realization is a structure*

$$\mathcal{R} = (\mathcal{A}_{\mathcal{R}}, \Pi_{GR}^{\mathcal{R}}, \Pi_{QM}^{\mathcal{R}}, E_{\mathcal{R}})$$

where $\mathcal{A}_{\mathcal{R}}$ is the realized quotient architecture, $\Pi_{GR}^{\mathcal{R}}$ and $\Pi_{QM}^{\mathcal{R}}$ are the GR-type and QM-type projections, and $E_{\mathcal{R}}$ is the explicit admissibility envelope.

Definition 12.2 (Realization certificate). *A realization certificate for candidate $\mathcal{R}$ consists of:*

- (C1) *an anchor ledger: domain, boundary data, regime, regularity class, and admissibility envelope;*
- (C2) *a tensor ledger: invariant observables, constraint witnesses, predictive witnesses, and amplitude or record witnesses;*
- (C3) *a skin ledger: gauge, coordinate, basis, representation, regulator, and ordering freedoms proved inert;*
- (C4) *a GR projection: coisotropic standing, characteristic redescription, first-class closure, and multiplier-type presentation or an admissibility-equivalent substitute;*
- (C5) *a QM projection: Hilbert/Schrödinger predictive interior or admissibility-equivalent structure;*
- (C6) *a boundary projection: Bell/record or measurement-output quotient well-definedness;*
- (C7) *a transport certificate: maps between regimes preserving quotient identity and admissible continuation;*
- (C8) *a matching ledger: regimes in which standard GR, QFT, and quantum-mechanical outputs are recovered within declared error envelopes.*

What supplying a certificate means. The C1–C8 entries are verified properties of one candidate $\mathcal{R}$ in its declared envelope $E_{\mathcal{R}}$. In particular, C3 requires a proof that each claimed freedom is inert; C4–C6 require lawful, well-defined projections of that candidate; C7 requires actual maps preserving the stated quotient identity and continuation; and C8 identifies the regimes and error bounds for the claimed recovery. Filling the ledger headings without those verifications does not supply a certificate.

For C8, the candidate must identify the outputs, regimes and error bounds for each recovery claim, and record exclusions and outstanding obligations explicitly. No entry establishes an unverified QFT extension. Whether a narrower candidate supplies all the data required by the certificate target must be checked against that target; keeping a matching ledger is not sufficient by itself. The same declared envelope governs all eight entries, and an outstanding claim within it cannot be erased by calling it skin or by silently narrowing the target.

The converse below concerns candidates claiming this C1–C8 realization target. Preserving GR-type and QM-type content in some narrower regimes does not, by itself, establish every QFT or boundary matching claim in a broader target. A candidate declaring a narrower recovery target must be assessed in that declared scope; it has not supplied the broader certificate. Thus the classification of a missing entry is relative to the target claimed, and does not pronounce a different physical theory impossible.

Theorem 12.3 (Certificate sufficiency). *A candidate supplying the data of Definition 12.2 is not a primitive quantization of gravity but a constraint-native quantum-gravity realization. Conversely, a candidate claiming to preserve both GR-type and QM-type standing-bearing content must supply these data or be classified as skin, scope change, auxiliary load, quotient/continuation motion, or standing failure.*

Proof. The anchor and tensor ledgers fix the admissibility type of the candidate. The skin ledger prevents representational surplus from carrying undeclared standing. The GR projection supplies the forced constrained-Hamiltonian interior of Definition 8.1. The QM projection supplies the predictive quantum theorem object. The boundary projection supplies record/witness fixation. The verified transport maps preserve the quotient identities and continuations specified in C7. The matching ledger establishes recovery only for the outputs, regimes and error envelopes actually asserted in C8.

If these data are supplied, neither GR nor QM is used as the primitive target of the other; both are projections of $\mathcal{A}_{\mathcal{R}}$. Conversely, a candidate claiming the same C1–C8 target must fix its anchors, tensors, quotient identities, continuations and boundary witnesses, and verify its claimed matching regimes. If an item is missing relative to that target, Theorem 4.7 classifies the proposed difference as inert skin, declared scope change, auxiliary load, or quotient/continuation motion; an undeclared standing-bearing difference fails standing. For a verified joint realization, Theorem 11.7 further gives the fixed-pair map criterion and excludes independent same-scope authority. Neither result manufactures a missing certificate entry. $\square$

Remark 12.4 (No metric-first or Hilbert-first primacy). The certificate program does not make metric geometry primitive and then add quantum structure, nor make Hilbert/operator structure primitive and then add gravity. Both are role-specific projections of the constraint architecture.

13 Non-claims, scope limits, and attack map

13.1 Failure modes if GR-type structure is removed

The main theorem is easiest to audit by removing one structural clause at a time. Each removal produces a determinate failure mode rather than a non-gravitational same-scope admissible world.

Removed or weakened clause	Failure reopened	Local control result
Kernel admissibility	no licensed/unlicensed continuation boundary; physical construction becomes unassessable	Lemma 3.3
Standing	outputs cannot be reused as same-domain physical content	Lemma 3.4
Reference	later acts do not have a determinate same object to continue	Lemma 3.5
Irreversibility	failed construction can be retroactively reset as successful	Lemma 3.6
Quotient identity	representative, coordinate, basis, or gauge choice becomes physical by privilege	Theorem 4.4
AER- $\text{Obs}_{\min}$	Poisson evolution of observables depends on ambient extensions or fails to descend	Proposition 6.5
AER-Sep	quotient-transverse distinctions remain undetected by physical observables	Proposition 6.5
AER-Max	hidden nonzero presymplectic directions survive invisibly	Proposition 6.5; Appendix B
First-class closure	admissible redescription is not closed under continuation	Theorem 7.5
AER-P multiplier typing	presentation freedom carries untyped physical force	Theorem 7.8

Proposition 13.1 (Removed-condition collapse). *Within the AER comparison class, removing any GR-type clause required by Definition 8.1 produces representation dependence, hidden load, quotient motion, continuation failure, or scope change. It does not produce a same-scope admissible interior without gravity in the AASC sense.*

Proof. The table enumerates the relevant first failures. If the removed clause is part of the kernel, one of Lemmas 3.3–3.6 applies. If quotient identity is removed, Theorem 4.4 fails and physical identity depends on representatives. If an AER clause is removed, Proposition 6.5 identifies the corresponding loss of closure, separation, or maximality. If first-class closure or multiplier typing is removed, Theorems 7.5 and 7.8 fail, reopening continuation motion or untyped presentation load.

These are typed failures, not admissible non-GR same-scope realizations. □

13.2 Non-claims

This paper does not claim:

- (NC1) that Einstein’s field equations, numerical coupling constants, or cosmological boundary conditions are derived from no premises;
- (NC2) that all modified-gravity theories are false;
- (NC3) that effective quantization, semiclassical gravity, gravitational EFT, loop/spinfoam methods, string-like theories, or other quantum-gravity programs are invalid;
- (NC4) that Lorentzian geometry or curvature is unreal;
- (NC5) that GR and QM are isomorphic as mathematical categories;
- (NC6) that every formal system, regardless of target, must contain GR-like geometry.

The paper does claim:

- (C1) that an admissible physical interior in the AER comparison class must instantiate gravity in the AASC sense: GR-type constraint architecture;
- (C2) that gravity in the AASC sense is therefore forced as a structural role;
- (C3) that an undeclared second foundational authorizer for a fixed jointly realized GR/QM target has no standing, and a compatible direct map requires the physical-fibre criterion;
- (C4) that successful quantum gravity should be constraint-native and certificate-bearing.

13.3 Hostile-referee attack map

Objection	Audit response
“You are redefining gravity.”	The use of “gravity” is explicitly fixed by Definition 2.1. The definition is not arbitrary: it names the constrained-Hamiltonian role canonically realized by GR and forced in Theorems 6.3, 7.5, and 7.8.
“This does not derive Einstein’s equations.”	Correct. The claim is structural necessity of gravity in the AASC sense, not numerical derivation of all dynamics. See Definitions 2.1 and 8.1.
“Modified gravity is possible.”	Yes. If it preserves the GR-type constraint architecture, it is a different tensor/dynamics inside the architecture; if it changes anchors or witnesses, it is richer scope, not same-scope counterexample.

Objection	Audit response
“Quantum gravity programs exist.”	The paper does not deny them. It types them as restricted-scope, richer-scope, or constraint-native realizations rather than primitive GR-to-QM maps.
“Curvature is physically real.”	Curvature is valid tensor content in geometric projection. Proposition 8.5 denies primitive status, not physical validity.
“AER-Max is smuggling in reduction.”	AER-Max excludes hidden presymplectic remainders; Appendix B gives a model satisfying descent and separation while failing AER-Max.
“The fixed-domain closure is taxonomic.”	The normal-form image theorem and collapse theorem, Theorems 10.6 and 10.7, show exhaustion and collapse under the two-locus bridge.
“A selector can choose a representative.”	A selector is anchor, tensor, or skin by Theorem 4.7; as independent same-scope authority it is excluded by Corollary 4.8.
“Primitive quantization can still work.”	A compatible direct map obeys the physical-fibre criterion; independent authorizing force is separately excluded by the fixed-target kernel. Changed pairing or physical data must be declared. See Theorem 11.7.
“The target phenomenon is circularly defined.”	The kernel target contains no metric, constraints, lapse, shift, or first-class algebra. The anti-circularity burden is discharged by Theorem 3.10.
“The AER theorem is merely imported.”	The local AER proof is reproduced in Sections 5–7; Theorem 7.10 lists the proof chain.
“Non-Hamiltonian or dissipative systems evade the theorem.”	They may be coherent outside the theorem target. The result is explicitly relative to the AER comparison class; see Definition 5.1 and Remark 5.2.
“Projection equivalence is metaphorical.”	Theorem 11.4 constructs canonical role quotients and separates their native identification. The common carrier, physical target coverage, native witness matching and full continuation checks are explicit realization obligations.

14 Conclusion

Gravity is therefore not a sector added to physical theory, but the structural necessity that makes admissible physical theory possible. This claim is strictly internal to the AASC comparison class defined in this paper: it identifies the structural condition under which a well-typed physical interior exists, not a universal derivation of all gravitational phenomena.

The contribution of this paper is not to describe gravitational structure, but to demonstrate that such structure is necessary for the existence of an admissible physical interior through the forcing chain proved above (Theorems 7.10 and 8.2; see Sections 5–6). This converts gravity from a contingent modeling choice into a structural inevitability.

The resulting statement is stronger than an interpretation of GR and weaker than an unrestricted derivation of all gravitational physics from nothing. It is stronger because it explains why the gravitational role is not merely a historical formalism: it is the unique same-scope normal form of admissible physical interiority in the AER comparison class. It is weaker because it remains comparison-class relative: richer-scope theories may add anchors, tensors, boundary data, or transport rules, but then they are not counterexamples to the same fixed target.

The use of the term “gravity” in this paper is role-theoretic. It does not assume that gravity is identical to curvature or any specific field content. It identifies the unique admissibility-bearing constraint role realized canonically by GR: the role without which standing, quotient identity, observable completeness, and lawful continuation cannot be maintained in the physical interior.

The projection result is a further layer. For a supplied joint presentation, independently specified standing uses determine canonical role profiles. Structural Unification establishes the paired one-interior result for a determinate joint target [26]. The constructive capstone proves complete certificate-relative realization of its exact frozen gravity–quantum target [28]. The native comparison uses that target’s verified maps, endpoints and histories. A common source supplies a paired relation; it supplies a direct gravitational-to-quantum map exactly under the physical-fibre criterion. The gravity-necessity proof is established before these realization results are invoked.

The shortest summary is:

Gravity is the forced constraint role of admissible physical existence.

Without the AASC gravity role, the admissible physical interior cannot maintain standing, quotient identity, observable completeness, and lawful continuation. Gravity is therefore not merely something the interior happens to contain. It is part of what makes the interior possible.

Accordingly, gravity appears not as a postulate but as the fixed point of admissibility under identity-preserving continuation.

All uses of the term “gravity” in this paper refer to the role defined in Definition 2.1; no claim is made that gravity is identical to curvature or any specific field content.

A Dependency ledger and source map

This appendix records how the present manuscript uses the AASC corpus. The local Hamiltonian proof needed for the main gravitational theorem is included in Sections 5–7. The corpus is used to identify dependency lineage, terminology, and broader normal-form closure context.

Local proof dependencies

The following ledger records the hypotheses used in the in-manuscript proof. It adds no condition to the AER comparison class or to Theorem 8.2.

Step	Input used	Output established
Vertical distribution	Smooth quotient submersion	$\mathcal{D} = \ker d\pi$, involutivity and connected local fibre leaves.
Coisotropy	AER-Obs _{min} , including brackets of two zero observables	Conormal Hamiltonian directions are tangent to $\mathcal{C}$.
Characteristic capture	Zero-observable brackets and AER-Sep	$(TC)^{\perp\omega} \subseteq \mathcal{D}$.
Reverse inclusion	Quotient observables and AER-Max	$\mathcal{D} \subseteq \ker \omega_{\mathcal{C}}$.
Reduced symplectic form	Characteristic equality, closedness, local submersion charts and the single quotient bracket	Local symplectic descent and bracket identification.
First-class presentation	Derived coisotropy, the local normal form and regular ideal factorization	Constraint generators, characteristic span and first-class closure.
Hamiltonian presentation	Tangency from AER-P(i), characteristic equality and regular defining functions	Reduced-plus-multiplier decomposition.
Presentation comparison	AER-P(ii)–(iii)	Common reduced dynamics; vertical differences and inert constants.
Compatible local lifts	A chosen reduced h , local extensions and regular defining functions	Compatible multiplier families; no selection of the physical reduced law.
Constraint basis invariance	Two regular length- k defining tuples for the same surface	Invertible local basis change; unchanged characteristic span and Hamiltonian.
Gravity necessity	All preceding steps, the existing AER hypothesis and Definitions 2.1, 8.1	The constraint role of Theorem 8.2.

Corpus lineage

The entries below identify the lineage stated in the source manuscript. The local implications just listed are established by the proofs and stated external mathematical input in this paper. Later metric, quantum, certificate and field realizations use that result downstream; their completion is not a premise of it.

Corpus source	Relevant result	Use in this paper
Minimal Conditions for Admissible Construction	AASC kernel; no faithful lower generator; no same-act repair; standing/reference/irreversibility necessity	Re-proved in compressed target-specific form in Section 3; cited for corpus lineage.
The Structure of Admissibility; AMetric Boundary; Bivalence papers	AMetric boundary, bivalence, unique admissible interior, standing conservation	Used as background for why admissibility is fail-closed and boundary/non-transport language is legitimate.
A Fixed-Domain Exhaustion Theorem; Closure by Exhaustion; repair papers	fixed-domain quotient discipline; no second same-scope classifier; repair exhaustion; no fifth locus	Recast in Sections 4 and 10.
Anchor, Tensor, and Skin	anchor/tensor/skin role decomposition; no fourth role	Proved in target-specific form as Theorem 4.7.
Admissibility and the Common Structural Origin of Gauge, Constraint, and Quantum Structure [18]	AER rigidity: coisotropic standing, first-class closure and multiplier form; common role-classification lineage	Local AER proof is reproduced in Sections 5–7; common role classification informs Section 11, where native realization is a separate obligation.
On the Non-Quantizability of Gravity and Constraint-Induced Rigidity of Physical Structure [19]	primitive quantization obstruction and realization certificate program	Recast and shortened in Sections 11–12.

Remark A.1 (What is imported versus proved). The local AER rigidity, canonical profile and factorization proofs are included in this paper. The broader corpus also supplies positive target-specific realization results: Structural Unification establishes the paired one-interior result, and the Modules A–D capstone proves complete certificate-relative realization of its declared target. These imported conclusions retain the native scopes and proof dependencies of their sources. The fixed-target correspondence in Section 11 explicitly matches the occupied closed packet’s structured native endpoints, standing uses and clock-indexed histories to Theorem 11.4(ii). The construction and complete certificate remain results of the cited downstream papers.

Projection proof and native-target ledger

Theorem 11.4(i) constructs role profiles from a supplied joint carrier and independently specified uses; (ii) gives the native-target comparison conditions. The gravity-necessity theorem and these local quotient proofs do not use the later realization results as premises. Structural Unification supplies the one-interior specification; the later constructive modules populate the joint target, and the capstone closes its certificate. Their positive completed-target conclusion is cited at this later

stage. A paper-level citation back to gravity is distinct from using the joint-realization conclusion as its own proof premise.

Input or result	Exact use and scope
Joint carrier and uses	Supplied physical target, lawful identity, invariant role and joint uses; separate targets do not construct it.
Canonical role profiles	In-manuscript construction $R_X = (P/\approx_P)/E_X$; unique partial-use descent. Histories require domain saturation and output congruence.
Native identification	Supplied raw maps, full physical target coverage, role-use fibre inclusion, native witness matching, anchor lifting and full target-history realization. Set-profile bijectivity does not prove analytic equivalence.
Einstein dynamics [22]	Theorem 30.1 classifies a faithful metric projection under its native assumptions. It supplies no general joint map constructor here.
Schrödinger dynamics [23]	Theorems 8.3 and 36.1 distinguish the standing handoff from the supplied spatial projection. Proposition 36.4 pairs normal forms and leaves joint compatibility to a further construction.
Physical quotient source [24]	Definitions R.1/R.5 and Theorem R.7 classify predictive, record and gravitational roles. Their role-triple maps are not the native common-source maps required here.
Structural Unification [26]	Theorems 5.1, 6.5 and 48.1 establish the same-interior paired result for the declared target bearing both kinds of standing.
Module C [25]	Theorems 24.7 and 27.1–27.5 construct a nonempty common carrier and simultaneous metric, full-ray quantum, record and compatibility processes in the certified local envelope.
Module D and capstone [27, 28]	Source-locked tensor and joint-skin completion, compatible occupation of all eight roles, and complete certificate-relative realization of the declared frozen target (capstone Theorems 47.4 and 70.5). This is a positive realization conclusion.

Closed-packet correspondence checks

The following checks use the completed closed-packet target and its predeclared native ingredients. Certificate-role occupation and physical state/history coverage have distinct proof sources.

Interface clause	Construction or proof used
Carrier and endpoints	Module C Definitions 24.1–24.5 give P_* and four native outputs. Capstone Definition 45.1 retains the AER object in the GR occupant and the record/compatibility occupants alongside the quantum process.

Interface clause	Construction or proof used
P1: anchors	Capstone Definitions 23.2 and 45.1 fix the envelope; Lemma 46.1 distinguishes new target parameters. Inside the fixed clock presentation these anchors are preserved. Module C Theorem 10.5 and Proposition 15.4 transport the whole correspondence between lawful clock presentations, with their durations.
P2–P3: standing uses	Module C Theorems 13.3, 13.6, 13.8 and 13.10 independently identify the complete native role profiles. The fixed AER object is carried explicitly; clock and failure context is retained. Capstone Definitions 25.1 and 45.1 fix the witness interfaces before their occupation.
P4: lawful identity	Module C Definition 13.11 and Theorem 15.2 give the role quotients and lawful projection functors. Module D Theorems 5.22 and 5.24 supply the distinct complete joint probe and lawful joint-skin kernel.
Coverage and P5	The explicit inverse lift and α_t in Section 11, using Module C Proposition 24.6, cover the declared local native states and segments. The retained clock makes their domains saturated; identities and compositions follow from the flow laws.
Joint compatibility	Capstone Lemma 47.3 and Theorem 47.4 verify the shared target and incidence diagram; Theorem 70.5 supplies complete certificate-relative realization. The local correspondence does not infer state/history coverage from the role-assignment bijection.

B Minimality witnesses

This appendix records concrete witnesses showing that the AER clauses are not decorative. The most important witness is the failure of AER-Max: observable Poisson descent and quotient separation do not by themselves force exact characteristic identification.

Proposition B.1 (Incomplete presymplectic quotient witness). *There is a regular local model satisfying AER-Obs_{min} and AER-Sep in which $\mathcal{C}$ is coisotropic but $\ker(\omega_{\mathcal{C}}) \subsetneq \mathcal{D}$. Hence AER-Max is a genuinely additional observable-completeness condition.*

Proof. Let

$$S = \mathbb{R}_{(q_1, p_1, q_2, p_2)}^4, \quad \omega = dq_1 \wedge dp_1 + dq_2 \wedge dp_2,$$

and let $\mathcal{C} = \{p_1 = 0\}$. Then $\mathcal{C}$ is coisotropic and

$$\ker(\omega_{\mathcal{C}}) = \text{span}\{\partial_{q_1}\}.$$

Define

$$\pi : \mathcal{C} \rightarrow \mathbb{R}, \quad \pi(q_1, q_2, p_2) = q_2.$$

Then

$$\mathcal{D} = \ker(d\pi) = \text{span}\{\partial_{q_1}, \partial_{p_2}\},$$

so $\ker(\omega_{\mathcal{C}}) \subsetneq \mathcal{D}$. Physical observables are functions of q_2 . Their differentials span $\text{Ann}(\mathcal{D}) = \text{span}\{dq_2\}$, so AER-Sep holds for this quotient. To check extension independence rather than only

the bracket of preferred representatives, write arbitrary extensions locally as $\overline{O} = F(q_2) + p_1 A$ and $\overline{P} = G(q_2) + p_1 B$, using the regular defining function p_1 . The bracket of $F(q_2)$ and $G(q_2)$ vanishes. Each remaining term either contains a factor p_1 or contains $\{p_1, F(q_2)\}$ or $\{p_1, G(q_2)\}$, which vanish. Hence every such bracket restricts to zero on $\mathcal{C}$, independently of extensions, and the zero bracket descends to the quotient. But $\partial_{p_2} \notin \ker(\omega_{\mathcal{C}})$ and every physical observable annihilates it. Thus AER-Max fails exactly by leaving a hidden nonzero presymplectic class invisible to all physical observables. $\square$

Corollary B.2 (Minimal observable completion). *Over AER-Obs_{min} and AER-Sep, any condition sufficient to force $\mathcal{D} = \ker(\omega_{\mathcal{C}})$ entails the content of AER-Max. No strict weakening of observable completeness can force the full GR-type normal form.*

Proof. Assume exact characteristic identification. If $v \in T_x \mathcal{C}$ and $v \notin \ker(\omega_{\mathcal{C},x}) = \mathcal{D}_x$, then by finite-dimensional linear algebra there exists $\alpha \in \text{Ann}(\mathcal{D}_x)$ with $\alpha(v) \neq 0$. AER-Sep says $\text{Ann}(\mathcal{D}_x)$ is spanned by differentials of physical observables, so some physical observable has $dO_x(v) \neq 0$. This is AER-Max. Proposition B.1 shows that without it exact identification can fail. $\square$

Second-class standing fails the zero-observable clause. The zero-observable part of AER-Obs_{min} already excludes a second-class standing surface relative to the stated ambient bracket. For example, take $S = \mathbb{R}^4_{(q_1, p_1, q_2, p_2)}$ with $\omega = dq_1 \wedge dp_1 + dq_2 \wedge dp_2$, let $\mathcal{C} = \{q_1 = p_1 = 0\}$, and use $\pi = \text{id}_{\mathcal{C}}$. The induced $\omega_{\mathcal{C}} = dq_2 \wedge dp_2$ is nondegenerate; all smooth quotient functions give AER-Sep and AER-Max. Yet q_1 and p_1 are both extensions of the zero restricted observable, whereas $\{q_1, p_1\}|_{\mathcal{C}} = -1$ with the sign convention used here. Their zero extensions have bracket zero. Thus extension independence fails. More generally, applying AER-Obs_{min} to all pairs of functions vanishing on $\mathcal{C}$ forces their restricted brackets to vanish; on a regular symplectic carrier this is the coisotropicity condition. This example isolates failure of Obs_{min}; it does not require a failure of AER-Max.

Solving second-class constraints first, or replacing the bracket by a Dirac bracket, changes the ambient presentation entering this interface. Such a construction must be checked with its own carrier and observable bracket. Nor does the example supply a separate failure witness for AER-Sep: all smooth functions on an ordinary smooth quotient already separate points and span cotangent directions. A restricted observable algebra or a different quotient category would change the stated comparison class. The witness above isolates Obs_{min}, while Proposition B.1 separately isolates Max.

Proposition B.3 (Ordering and measure witnesses). *If an operator ordering, regularization, path-integral measure, contour, graph/refinement rule, or canonical representative changes physical states, observables, amplitudes, records, or closure, then it is standing-bearing tensor or anchor load. If it changes none of these, it is skin. It cannot remain an undeclared same-domain convention while carrying physical force.*

Proof. Apply Theorem 4.7. If the choice changes well-typedness, domain, boundary, or regime, it is anchor. If it changes output-bearing invariant content, it is tensor. If it changes neither, it is skin. An undeclared fourth use is hidden load or selector authority and is excluded by Corollary 4.8. $\square$

C Symbol index

Symbol	Meaning
K	AASC kernel (Adm, Stand, Ref, Irr).
S, ω	Symplectic state carrier used in the regular-stratum Hamiltonian specialization.
$\mathcal{C}$	Standing surface, the admissible state locus.
$\mathcal{G}$	Local pseudogroup of admissible redescription.
$\pi : \mathcal{C} \rightarrow S_{\text{phys}}$	Quotient projection from standing surface to physical quotient.
$\mathcal{D}$	Redescription distribution $\ker(d\pi)$.
$\omega_{\mathcal{C}}$	Pullback of ω to $\mathcal{C}$.
ϕ_a	Local first-class constraint generators defining $\mathcal{C}$.
H_{α}	Presentation Hamiltonian indexed by presentation parameter α .
h_{phys}	Reduced physical Hamiltonian on S_{phys} , when present.
$\mathcal{A}$	Architecture of the declared joint presentation, with its lawful physical quotient.
G, Q	GR-type and QM-type theorem objects.
Π_{GR}, Π_{QM}	Projections from $\mathcal{A}$ to GR-type and QM-type theorem objects.

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